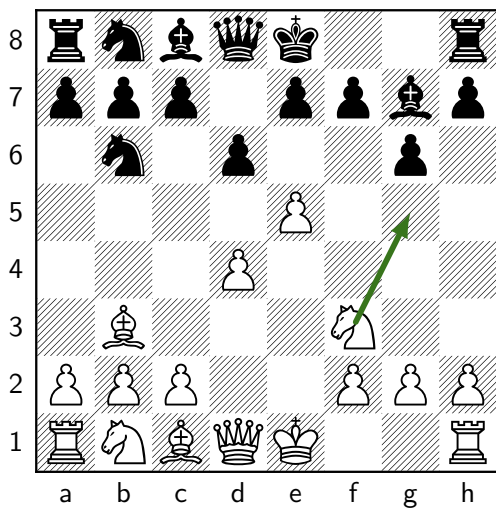


Melanie-BOTS (CCRL engines) — Alekhine Defense

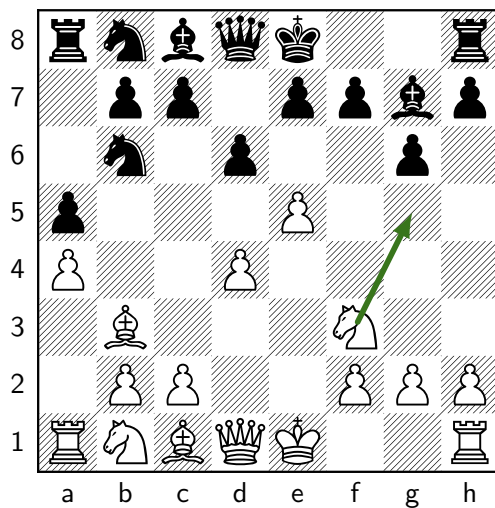
Melanie (Deep_Vision)

Abstract

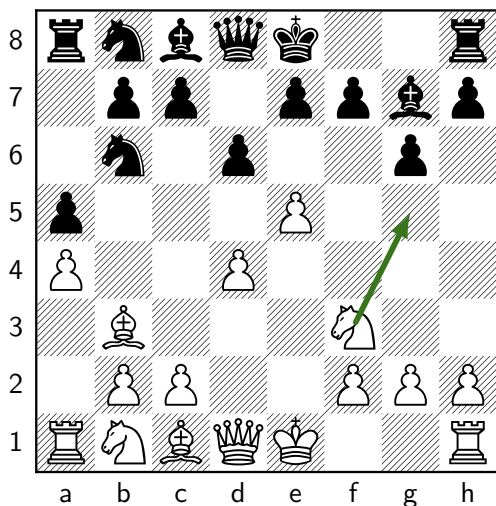
Every move below was produced by relational analogy to named master games (human and/or top-engine – the exact source is named per move). Each figure shows our position with Melanie’s move (a) and the top retrieved analog positions (b–d) with the master move played there; the caption names the exact source and the piece-role correspondence under μ – cross-type maps (a rook playing a queen’s role) are called out, as no material/eval engine explains a move that way.



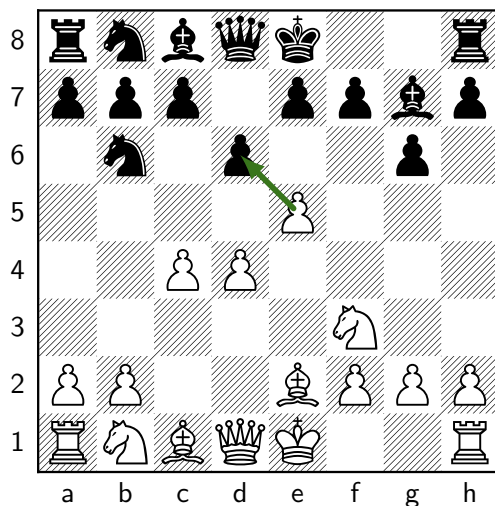
(a)



(b)

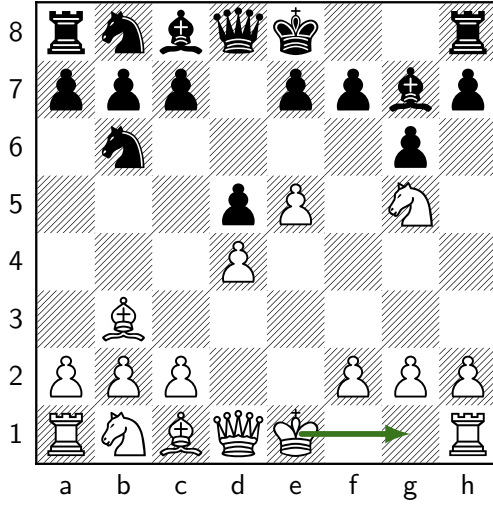


(c)

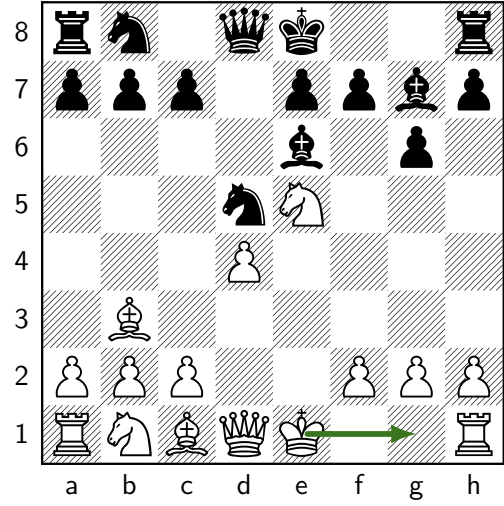


(d)

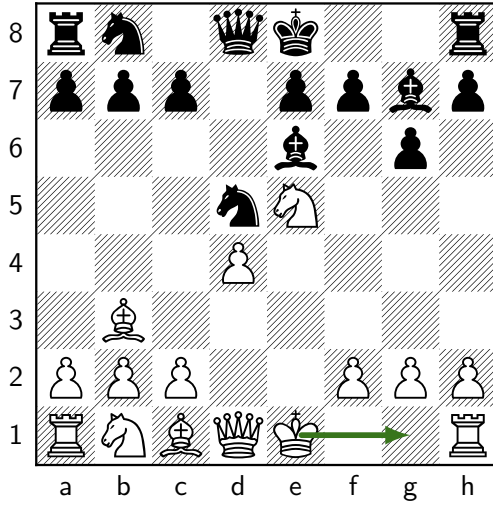
Figure 1: **Move 7, White: Ng5.** (a) Our position; Melanie plays Ng5 (f3g5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0002#1217 (ply 14) [structural distance $d=11$, QAP obj=83]: the master played Ng5; under μ our Nf3 plays the role of that game’s Nf3; cross-type roles: our Pd4 as their Ra1, our Ra1 as their Nb1; transferred move f3g5 (**endorses this move**). (c) Retrieved from ccr1_top_p0002#1218 (ply 14) [structural distance $d=11$, QAP obj=83]: the master played Ng5; under μ our Nf3 plays the role of that game’s Nf3; cross-type roles: our Pd4 as their Ra1, our Ra1 as their Nb1; transferred move f3g5 (**endorses this move**). (d) Retrieved from ccr1_top_p0001#637 (ply 12) [structural distance $d=37$, QAP obj=74]: the master played exd6; under μ our Pe5 plays the role of that game’s Pe5; cross-type roles: our pa7 as their Ra1, our Ra1 as their Nb1; transferred move e5d6. Synthesis: 2 retrieved analogs transferred to Ng5 (summed QAP objective 165, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



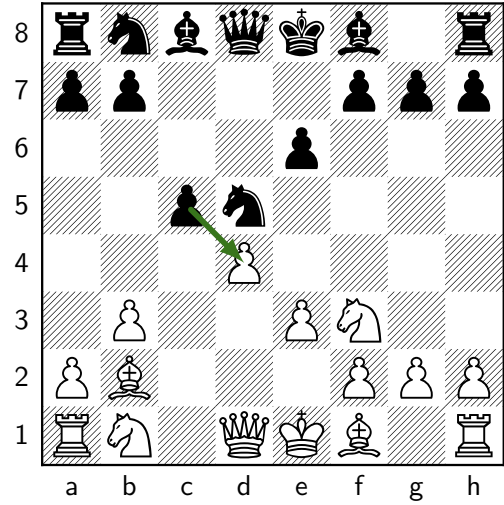
(a)



(b)

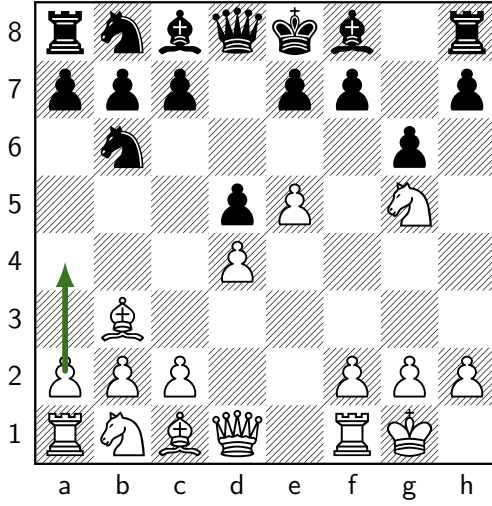


(c)

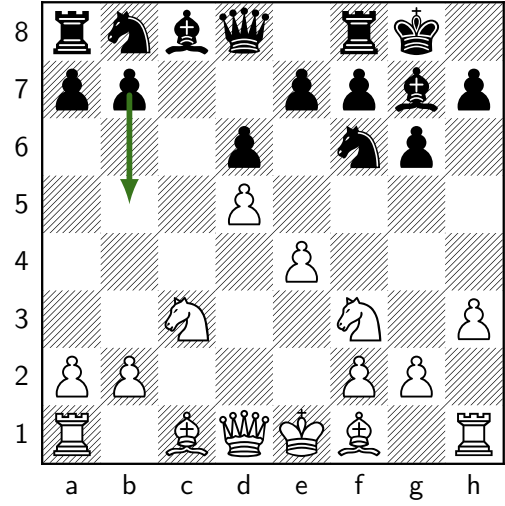


(d)

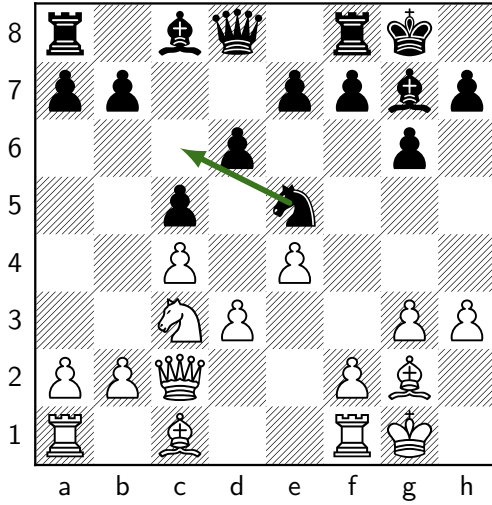
Figure 2: **Move 8, White: 0-0.** (a) Our position; Melanie plays 0-0 (e1g1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `cctl_top_p0002#3952` (ply 14) [structural distance $d=46$, QAP obj=30]: the master played 0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *Pd4* as their *Nb1*, our *Qd1* as their *Bc1*; transferred move e1g1 (**endorses this move**). (c) Retrieved from `cctl_top_p0002#3953` (ply 14) [structural distance $d=46$, QAP obj=30]: the master played 0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *Pd4* as their *Nb1*, our *Qd1* as their *Bc1*; transferred move e1g1 (**endorses this move**). (d) Retrieved from `cctl_top_p0003#1125` (ply 13) [structural distance $d=46$, QAP obj=28]: the master played *cxd4*; under μ our *Ng5* plays the role of that game’s *Pc4* (a **cross-type** correspondence); cross-type roles: our *Ph2* as their *Bc8*, our *Pf2* as their *Rh8*; transferred move g5e6. Synthesis: 2 retrieved analogs transferred to 0-0 (summed QAP objective 60, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



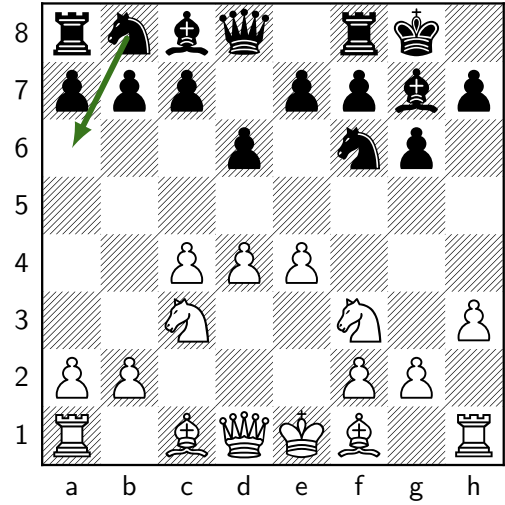
(a)



(b)

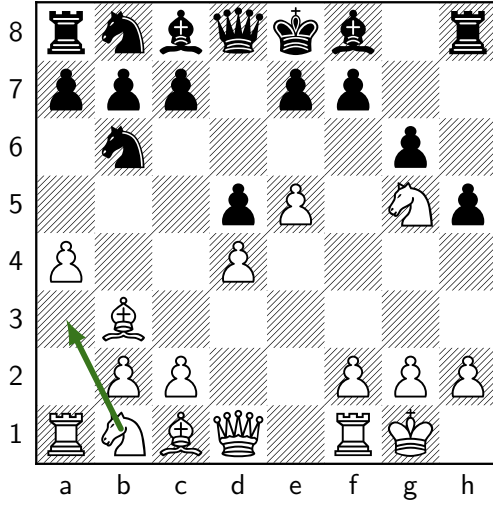


(c)

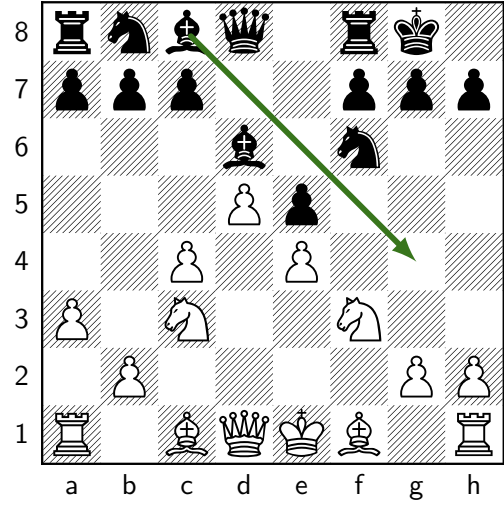


(d)

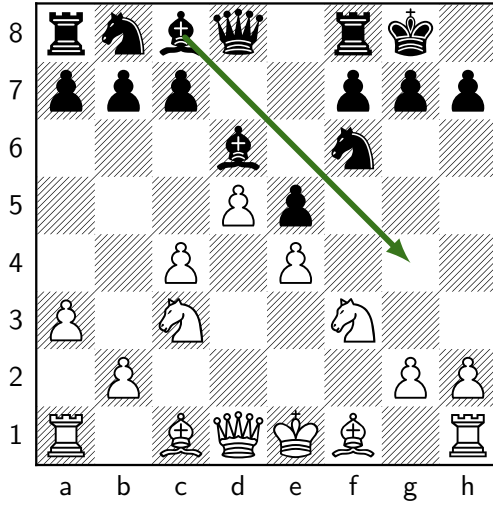
Figure 3: **Move 9, White: a4.** (a) Our position; Melanie plays **a4** (a2a4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#3469` (ply 15) [structural distance $d=52$, QAP obj=61]: the master played **b5**; under μ our *Pa2* plays the role of that game’s *Pb2*; cross-type roles: our *pb7* as their *Ra8*, our *pg6* as their *Nb8*; transferred move **a2a4** (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#1749` (ply 23) [structural distance $d=63$, QAP obj=61]: the master played **Nc6**; under μ our *Ng5* plays the role of that game’s *Ne4*; cross-type roles: our *Pb2* as their *Ra8*, our *Ra1* as their *Pa7*; transferred move **g5e4**. (d) Retrieved from `ccrl_top_p0000#1166` (ply 11) [structural distance $d=58$, QAP obj=34]: the master played **Na6**; under μ our *Pe5* plays the role of that game’s *Nb1* (a **cross-type** correspondence); cross-type roles: our *Pc2* as their *Qd8*, our *Nb1* as their *Pa7*; transferred move **e5e6**. Synthesis: 1 retrieved analog transferred to **a4** (summed QAP objective 61, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



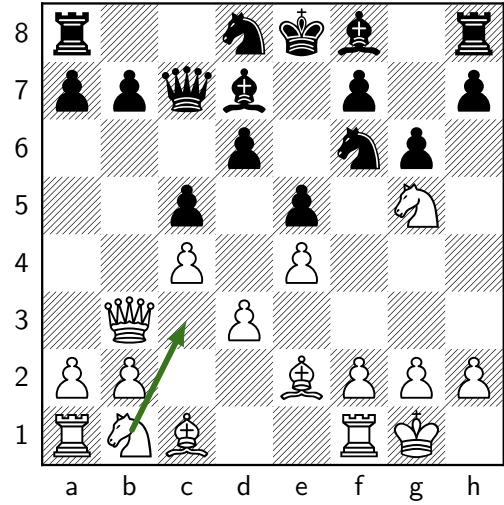
(a)



(b)

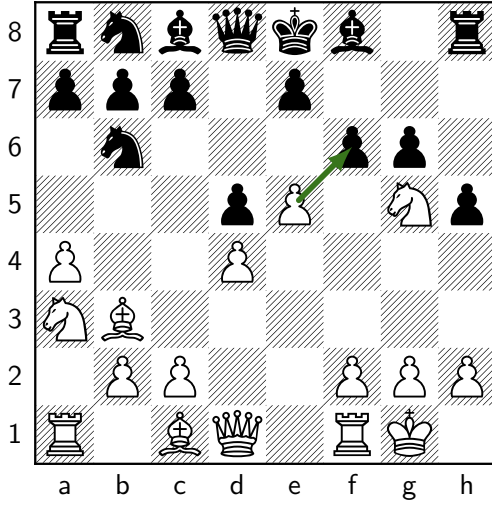


(c)

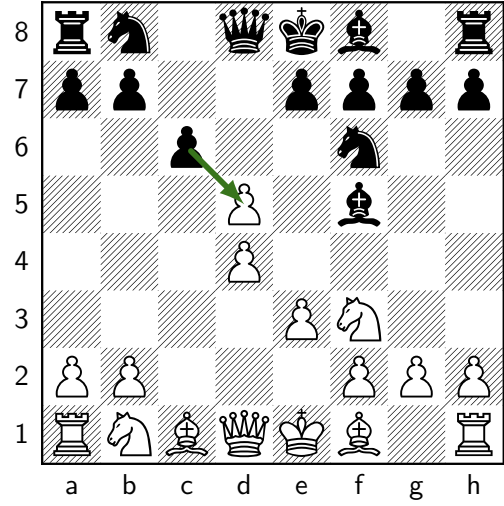


(d)

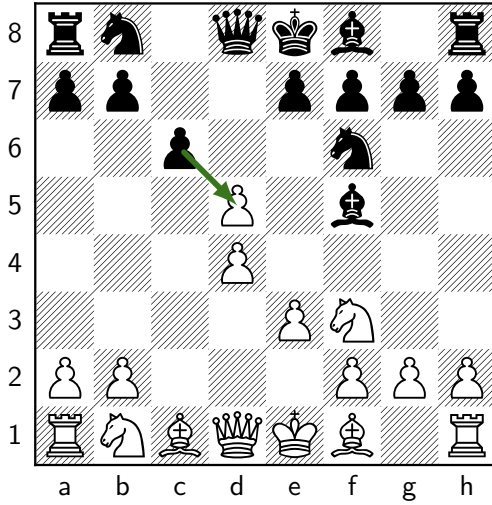
Figure 4: **Move 10, White: Na3**. (a) Our position; Melanie plays Na3 (b1a3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#2600` (ply 17) [structural distance $d=62$, QAP obj=75]: the master played Bg4; under μ our Nb1 plays the role of that game’s Bc1 (a **cross-type** correspondence); cross-type roles: our bc8 as their Ra8, our Ra1 as their Nb8; transferred move b1a3 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#2601` (ply 17) [structural distance $d=62$, QAP obj=75]: the master played Bg4; under μ our Nb1 plays the role of that game’s Bc1 (a **cross-type** correspondence); cross-type roles: our bc8 as their Ra8, our Ra1 as their Nb8; transferred move b1a3 (**endorses this move**). (d) Retrieved from `ccrl_top_p0003#1396` (ply 18) [structural distance $d=62$, QAP obj=73]: the master played Nc3; under μ our Ng5 plays the role of that game’s Nb1; cross-type roles: our pb7 as their Bc1, our Nb1 as their Pa2; transferred move g5f3. Synthesis: 2 retrieved analogs transferred to Na3 (summed QAP objective 150, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



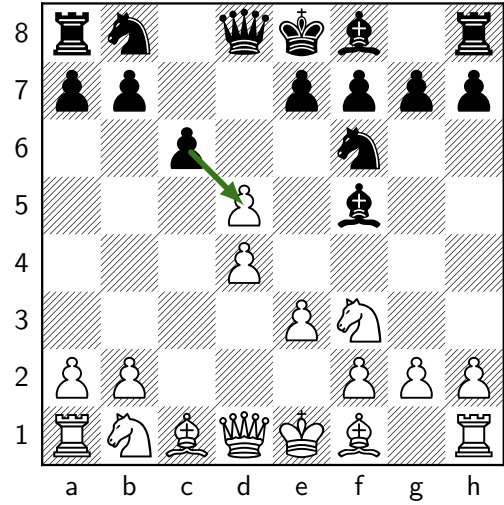
(a)



(b)

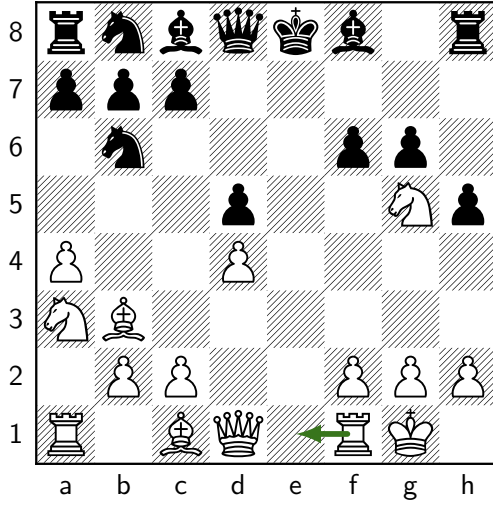


(c)

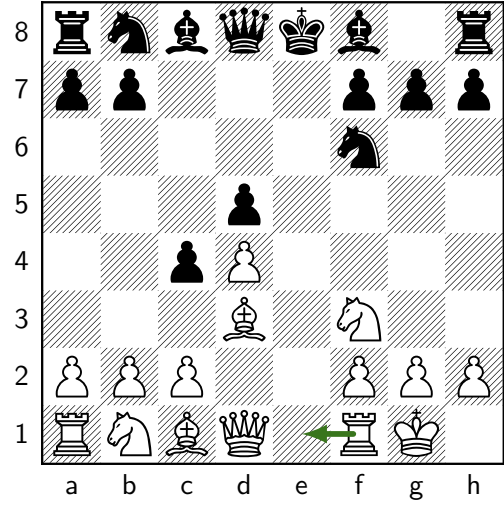


(d)

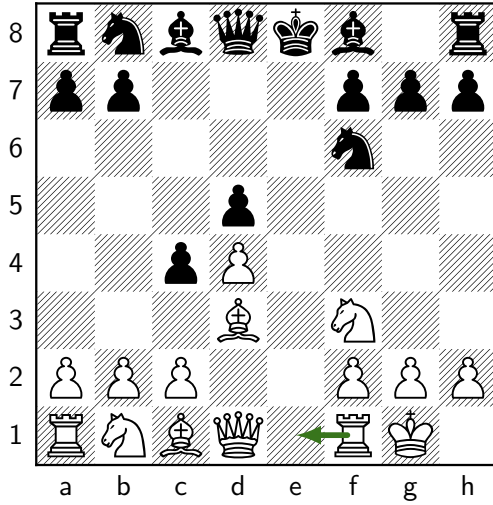
Figure 5: **Move 11, White: exf6.** (a) Our position; Melanie plays **exf6** (**e5f6**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top.p0001#762` (ply 9) [structural distance $d=59$, QAP obj=23]: the master played **cxd5**; under μ our *Pe5* plays the role of that game’s *Pc3*; cross-type roles: our *nb8* as their *Ra8*, our *Bc1* as their *Nb8*; transferred move **e5f6** (**endorses this move**). (c) Retrieved from `ccr1_top.p0001#763` (ply 9) [structural distance $d=59$, QAP obj=23]: the master played **cxd5**; under μ our *Pe5* plays the role of that game’s *Pc3*; cross-type roles: our *nb8* as their *Ra8*, our *Bc1* as their *Nb8*; transferred move **e5f6** (**endorses this move**). (d) Retrieved from `ccr1_top.p0003#126` (ply 9) [structural distance $d=59$, QAP obj=23]: the master played **cxd5**; under μ our *Pe5* plays the role of that game’s *Pc3*; cross-type roles: our *nb8* as their *Ra8*, our *Bc1* as their *Nb8*; transferred move **e5f6** (**endorses this move**). Synthesis: 5 retrieved analogs transferred to **exf6** (summed QAP objective 113, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



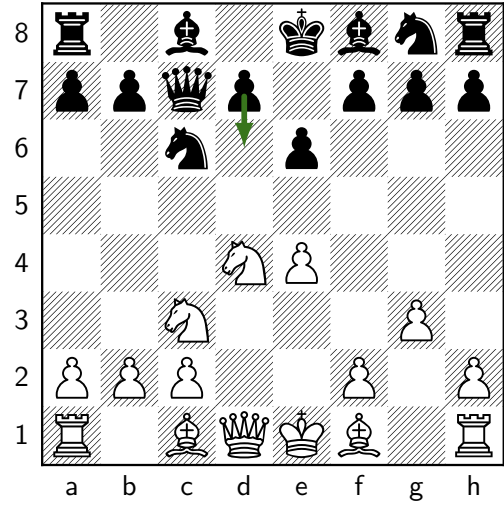
(a)



(b)

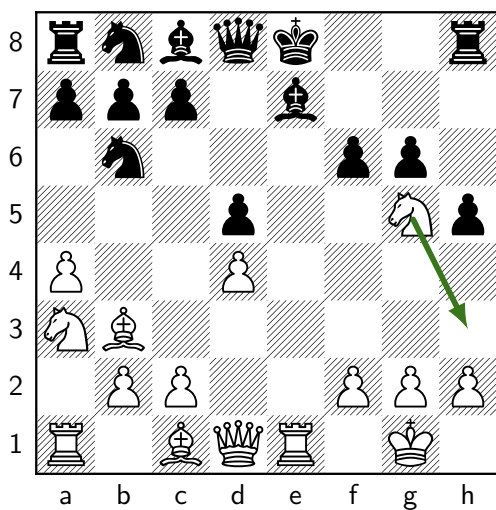


(c)

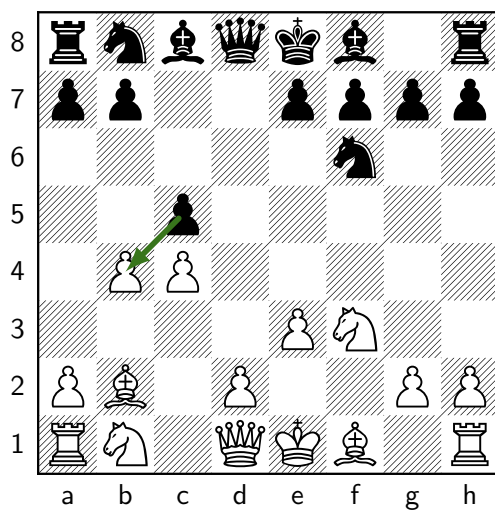


(d)

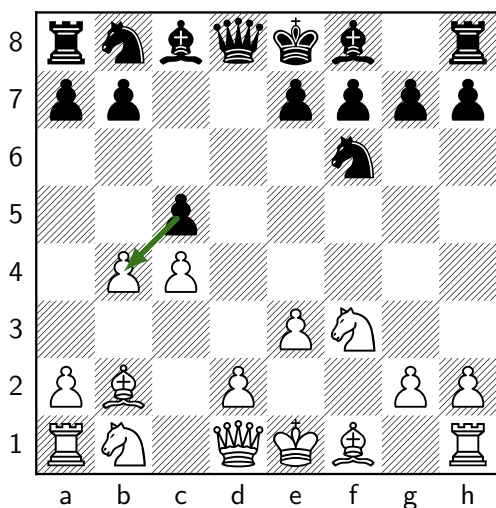
Figure 6: **Move 12, White: Re1+.** (a) Our position; Melanie plays Re1+ (f1e1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#1267` (ply 12) [structural distance $d=62$, QAP obj=25]: the master played Re1+; under μ our Rf1 plays the role of that game’s Rf1; cross-type roles: our Bc1 as their Ra1, our pg6 as their Nb1; transferred move f1e1 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#1268` (ply 12) [structural distance $d=62$, QAP obj=25]: the master played Re1+; under μ our Rf1 plays the role of that game’s Rf1; cross-type roles: our Bc1 as their Ra1, our pg6 as their Nb1; transferred move f1e1 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#1194` (ply 11) [structural distance $d=64$, QAP obj=14]: the master played d6; under μ our Rf1 plays the role of that game’s Pd2 (a **cross-type** correspondence); cross-type roles: our Qd1 as their Ra8, our Na3 as their Bc8; transferred move f1e1 (**endorses this move**). Synthesis: 8 retrieved analogs transferred to Re1+ (summed QAP objective 137, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



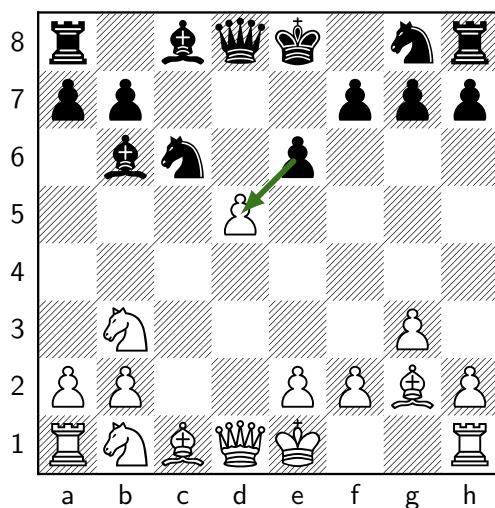
(a)



(b)

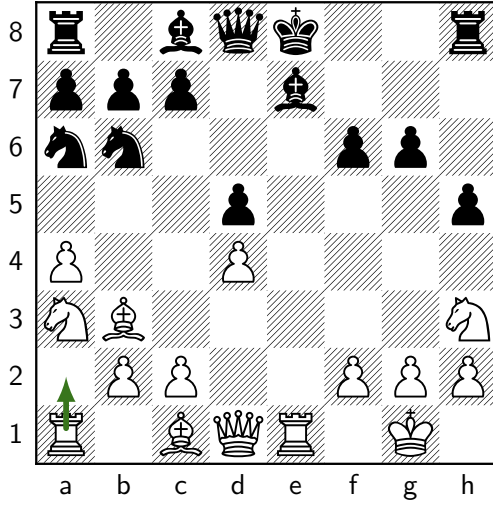


(c)

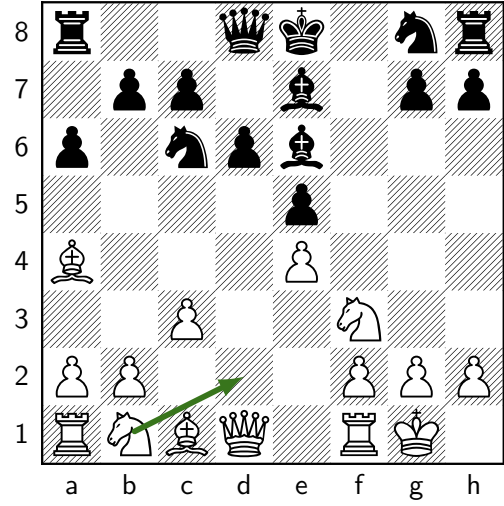


(d)

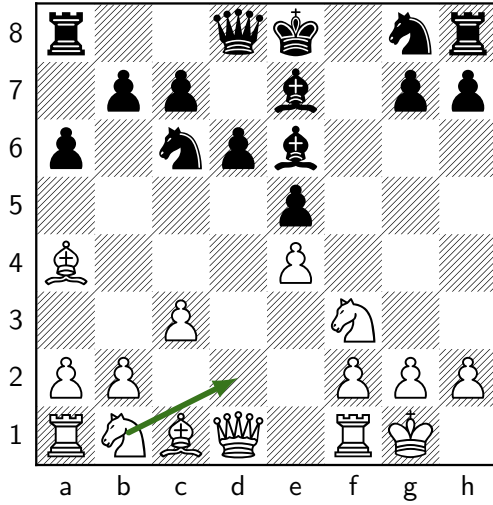
Figure 7: **Move 13, White: Nh3.** (a) Our position; Melanie plays Nh3 (g5h3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#3664` (ply 11) [structural distance $d=61$, QAP obj=24]: the master played `cxb4`; under μ our `Ng5` plays the role of that game’s `Pc4` (a **cross-type** correspondence); cross-type roles: our `Bb3` as their `Ra8`, our `Ra1` as their `Nb8`; transferred move `g5h3` (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#3665` (ply 11) [structural distance $d=61$, QAP obj=24]: the master played `cxb4`; under μ our `Ng5` plays the role of that game’s `Pc4` (a **cross-type** correspondence); cross-type roles: our `Bb3` as their `Ra8`, our `Ra1` as their `Nb8`; transferred move `g5h3` (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#3955` (ply 15) [structural distance $d=67$, QAP obj=62]: the master played `exd5`; under μ our `Na3` plays the role of that game’s `Pe3` (a **cross-type** correspondence); cross-type roles: our `Bc1` as their `Ra8`, our `Ra1` as their `Bc8`; transferred move `a3c4`. Synthesis: 2 retrieved analogs transferred to Nh3 (summed QAP objective 48, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



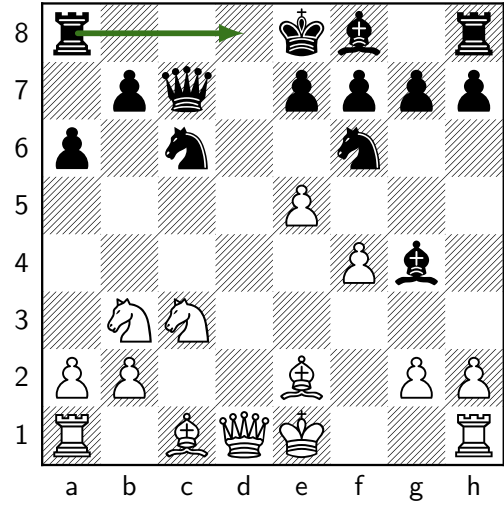
(a)



(b)

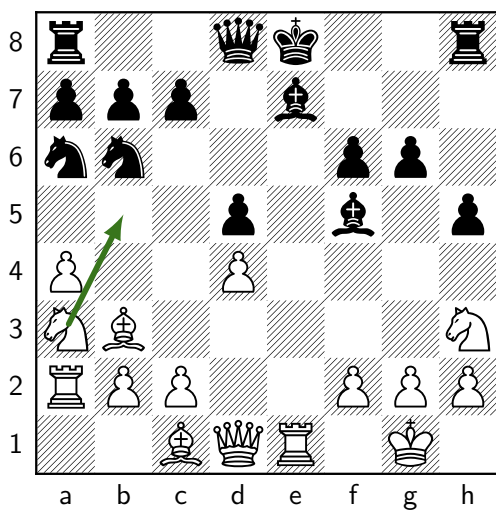


(c)

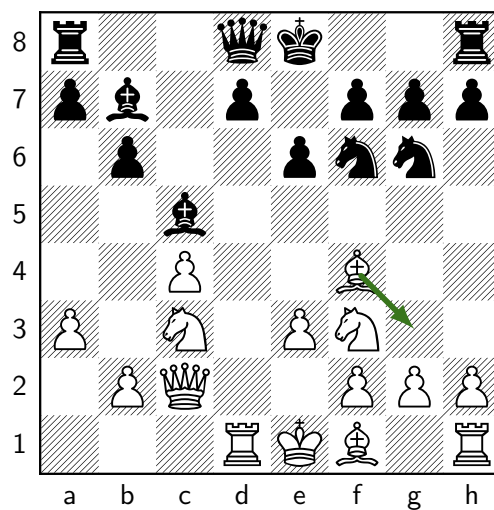


(d)

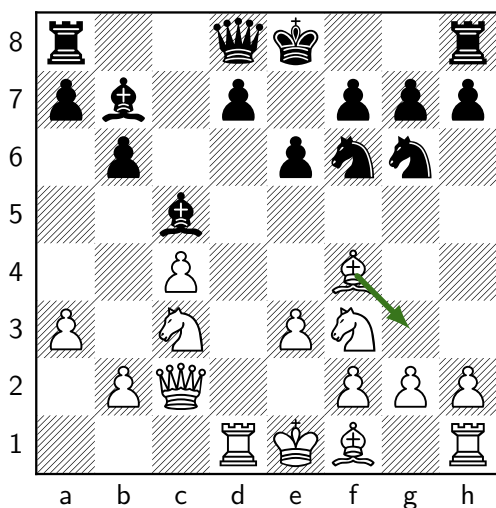
Figure 8: **Move 14, White: Ra2.** (a) Our position; Melanie plays Ra2 (a1a2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#1850` (ply 16) [structural distance $d=64$, QAP obj=32]: the master played Nbd2; under μ our Ra1 plays the role of that game’s Nb1 (a **cross-type** correspondence); cross-type roles: our Bb3 as their Ra1, our Na3 as their Bc1; transferred move a1a2 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#1851` (ply 16) [structural distance $d=64$, QAP obj=32]: the master played Nbd2; under μ our Ra1 plays the role of that game’s Nb1 (a **cross-type** correspondence); cross-type roles: our Bb3 as their Ra1, our Na3 as their Bc1; transferred move a1a2 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#1633` (ply 25) [structural distance $d=62$, QAP obj=20]: the master played Rd8; under μ our Ra1 plays the role of that game’s Ra1; cross-type roles: our Ph2 as their Bf8, our Bc1 as their Rh8; transferred move a1b1. Synthesis: 2 retrieved analogs transferred to Ra2 (summed QAP objective 64, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



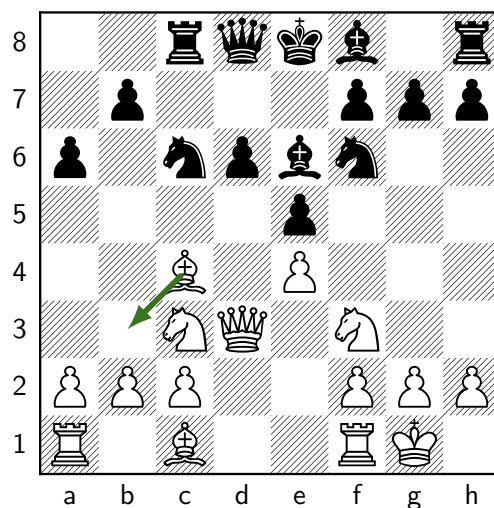
(a)



(b)

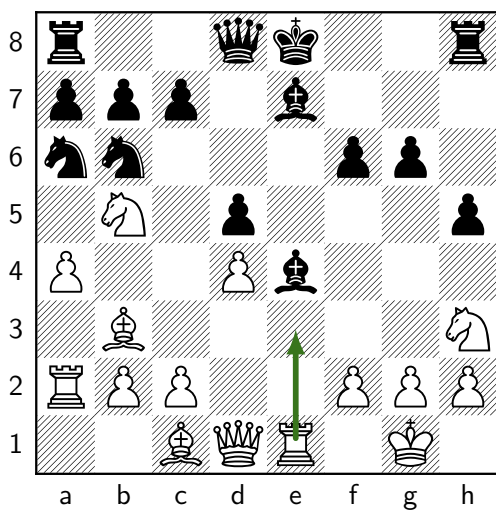


(c)

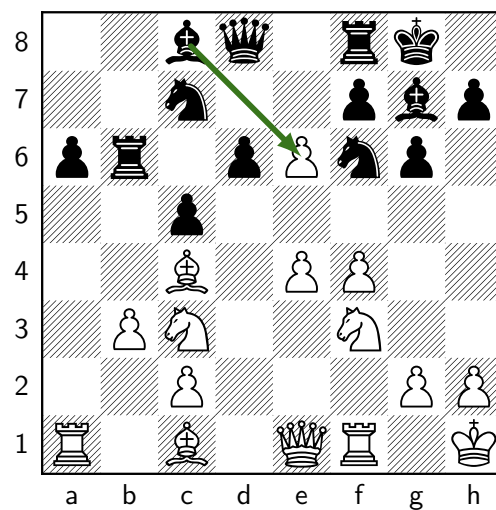


(d)

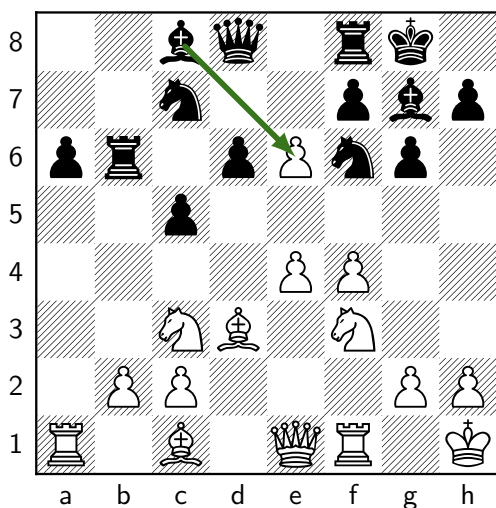
Figure 9: **Move 15, White: Nb5.** (a) Our position; Melanie plays Nb5 (a3b5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#456` (ply 20) [structural distance $d=70$, QAP obj=58]: the master played Bg3; under μ our Na3 plays the role of that game’s Bf4 (a **cross-type** correspondence); cross-type roles: our Qd1 as their Bf1, our Re1 as their Qc2; transferred move a3b5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#457` (ply 20) [structural distance $d=70$, QAP obj=58]: the master played Bg3; under μ our Na3 plays the role of that game’s Bf4 (a **cross-type** correspondence); cross-type roles: our Qd1 as their Bf1, our Re1 as their Qc2; transferred move a3b5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#1648` (ply 18) [structural distance $d=72$, QAP obj=50]: the master played Bb3; under μ our Nh3 plays the role of that game’s Bc4 (a **cross-type** correspondence); cross-type roles: our Bc1 as their Ra1, our Pd4 as their Bc1; transferred move h3f4. Synthesis: 2 retrieved analogs transferred to Nb5 (summed QAP objective 116, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



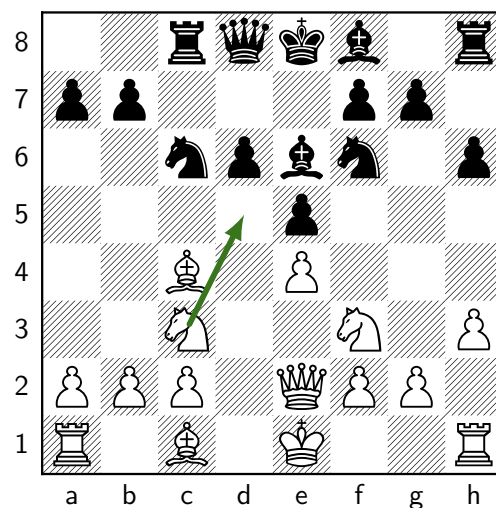
(a)



(b)

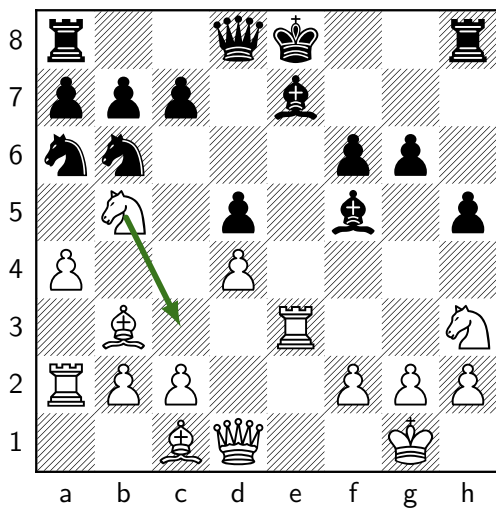


(c)

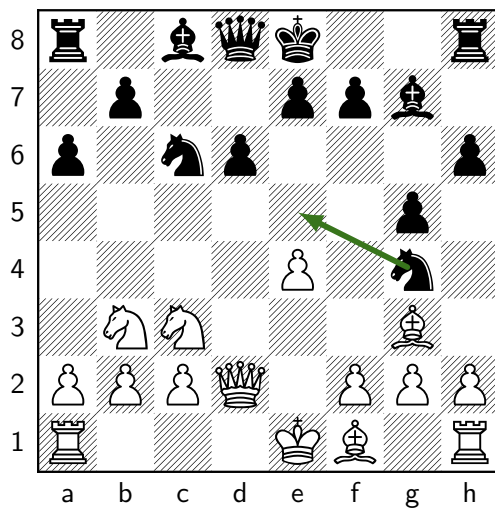


(d)

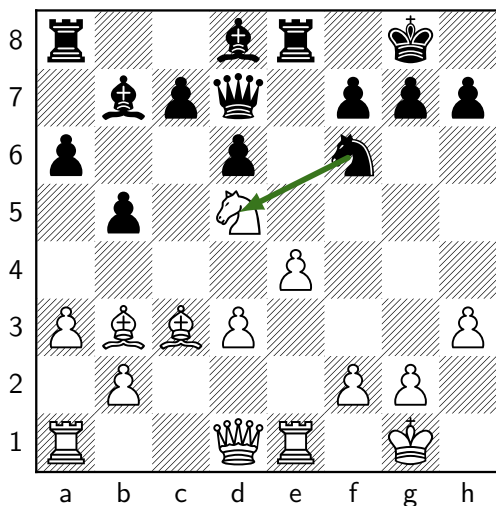
Figure 10: **Move 16, White: Re3.** (a) Our position; Melanie plays Re3 (e1e3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#1781` (ply 31) [structural distance $d=72$, QAP obj=67]: the master played Bxe6; under μ our Re1 plays the role of that game’s Bc1 (a **cross-type** correspondence); cross-type roles: our Ph2 as their Rf8, our Re1 as their Nc7; transferred move e1e3 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#752` (ply 31) [structural distance $d=74$, QAP obj=62]: the master played Bxe6; under μ our Re1 plays the role of that game’s Bc1 (a **cross-type** correspondence); cross-type roles: our Pg2 as their Qd8, our Nh3 as their Rf8; transferred move e1e3 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#3016` (ply 18) [structural distance $d=69$, QAP obj=31]: the master played Nd5; under μ our Bb3 plays the role of that game’s Nc3 (a **cross-type** correspondence); cross-type roles: our pg6 as their Ra1, our Qd1 as their Bc1; transferred move b3c4. Synthesis: 2 retrieved analogs transferred to Re3 (summed QAP objective 129, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



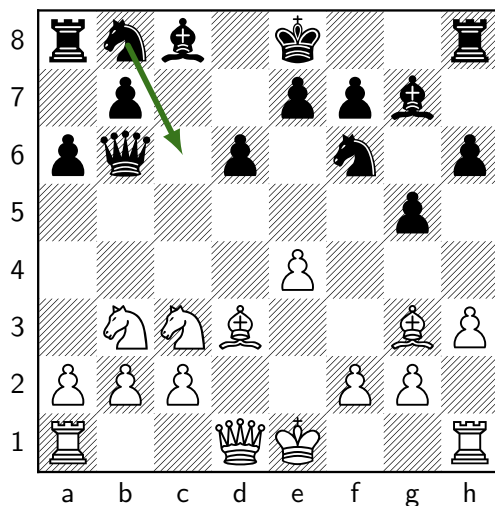
(a)



(b)

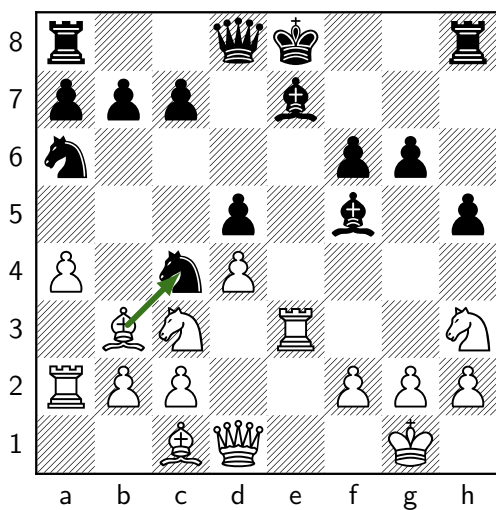


(c)

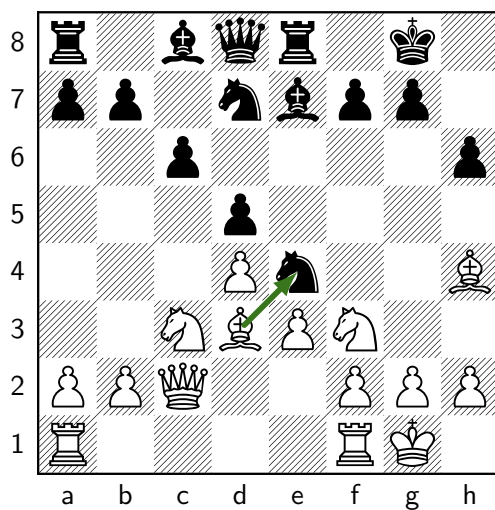


(d)

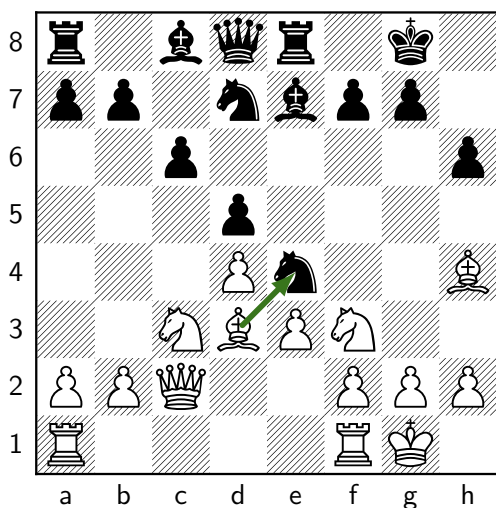
Figure 11: **Move 17, White: Nc3.** (a) Our position; Melanie plays Nc3 (b5c3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1989` (ply 21) [structural distance $d=76$, QAP obj=38]: the master played Nge5; under μ our Nb5 plays the role of that game’s Ng5; cross-type roles: our Ph2 as their Ra8, our Ra2 as their Bc8; transferred move b5c3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#1957` (ply 31) [structural distance $d=75$, QAP obj=50]: the master played Nxd5; under μ our Nh3 plays the role of that game’s Nf3; cross-type roles: our Pb2 as their Ra8, our Ph2 as their Bd8; transferred move h3g5. (d) Retrieved from `ccr1_top_p0002#2679` (ply 23) [structural distance $d=73$, QAP obj=44]: the master played Nc6; under μ our Pg2 plays the role of that game’s Nb1 (a **cross-type** correspondence); cross-type roles: our Pd4 as their Bc8, our Qd1 as their Rh8; transferred move g2g3. Synthesis: 1 retrieved analog transferred to Nc3 (summed QAP objective 38, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



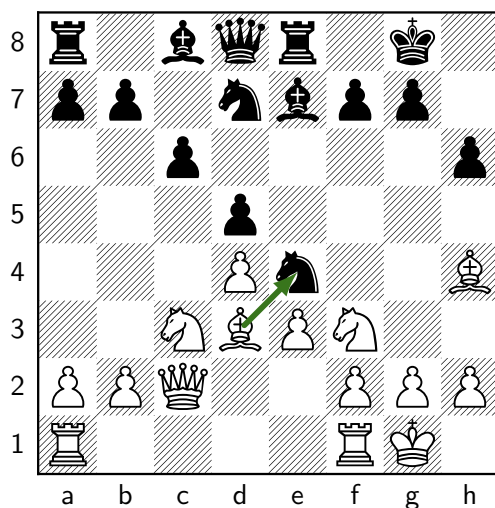
(a)



(b)

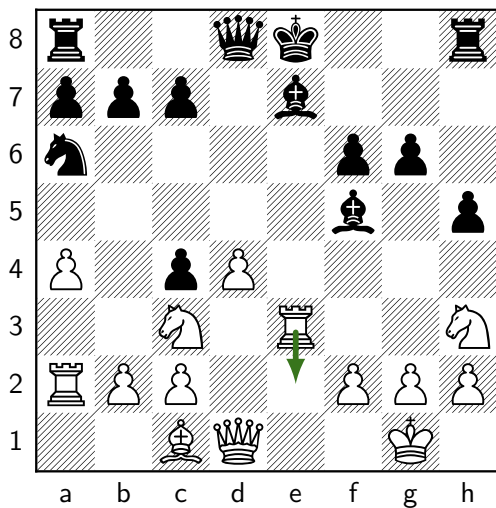


(c)

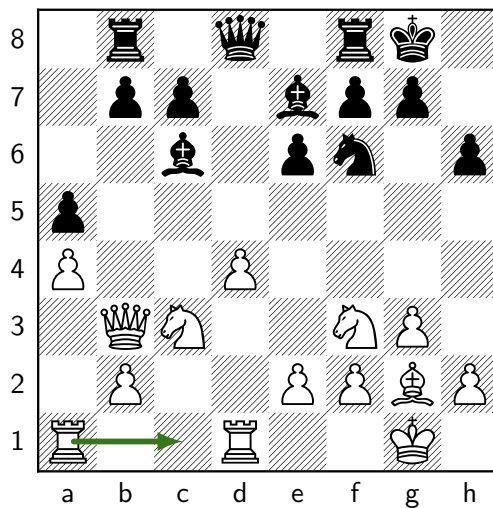


(d)

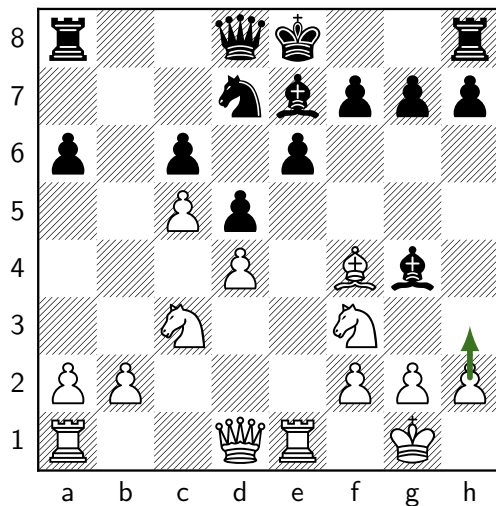
Figure 12: **Move 18, White: Bxc4.** (a) Our position; Melanie plays Bxc4 (b3c4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#702` (ply 22) [structural distance $d=96$, QAP obj=62]: the master played Bxe4; under μ our Bb3 plays the role of that game’s Bd3; cross-type roles: our Qd1 as their Ra1, our Nh3 as their Rf1; transferred move b3c4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3085` (ply 22) [structural distance $d=96$, QAP obj=62]: the master played Bxe4; under μ our Bb3 plays the role of that game’s Bd3; cross-type roles: our Qd1 as their Ra1, our Nh3 as their Rf1; transferred move b3c4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#1199` (ply 22) [structural distance $d=96$, QAP obj=62]: the master played Bxe4; under μ our Bb3 plays the role of that game’s Bd3; cross-type roles: our Qd1 as their Ra1, our Nh3 as their Rf1; transferred move b3c4 (**endorses this move**). Synthesis: 9 retrieved analogs transferred to Bxc4 (summed QAP objective 530, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



(a)

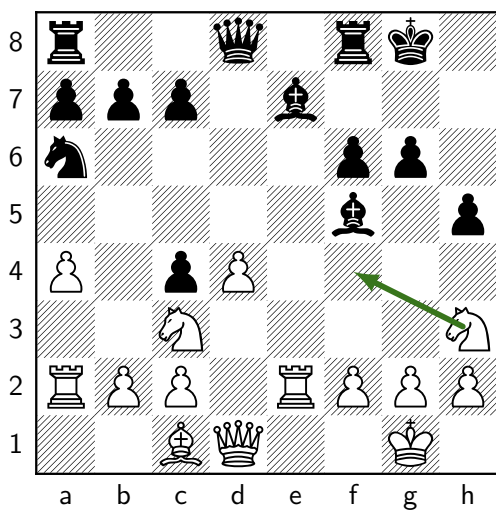


(b)

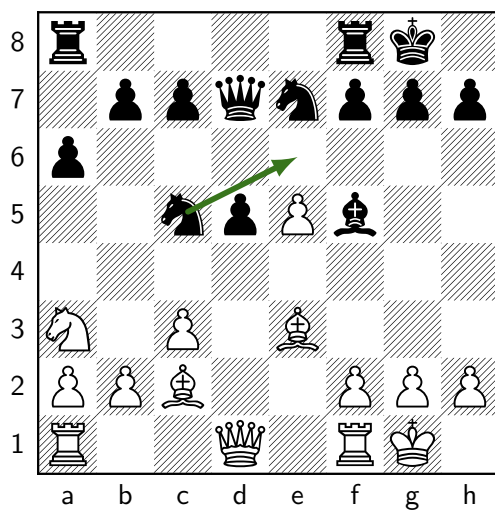


(c)

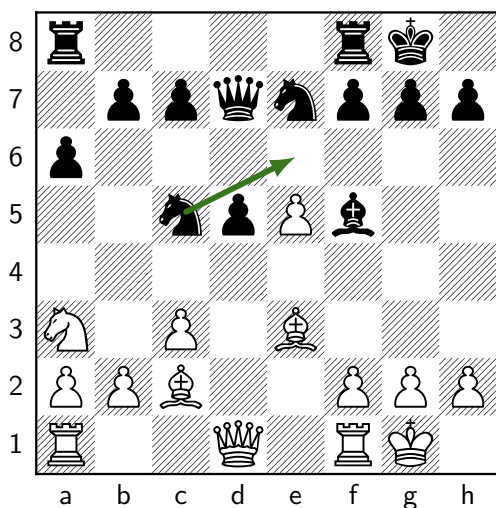
Figure 13: **Move 19, White: Re2.** (a) Our position; Melanie plays **Re2** (**e3e2**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#963` (ply 28) [structural distance $d=71$, QAP obj=33]: the master played **Rac1**; under μ our **Re3** plays the role of that game’s **Ra1**; cross-type roles: our **Qd1** as their **Rd1**, our **Bc1** as their **Pb2**; transferred move **e3e1**. (c) Retrieved from `ccr1_top_p0000#3201` (ply 24) [structural distance $d=74$, QAP obj=19]: the master played **h3**; under μ our **Pf2** plays the role of that game’s **Ph2**; cross-type roles: our **pa7** as their **Ra1**, our **Re3** as their **Qd1**; transferred move **f2f4**. Synthesis: no retrieved analog transferred to **Re2** – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



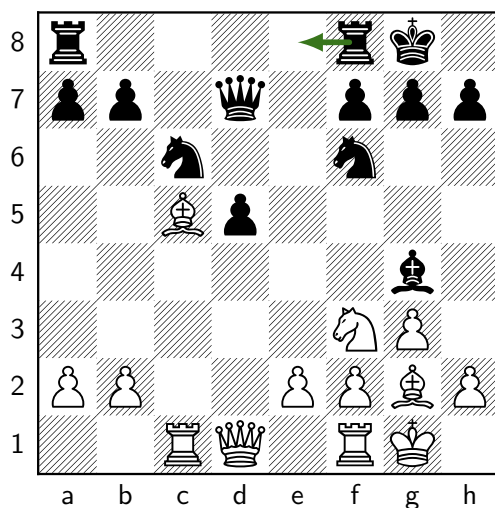
(a)



(b)

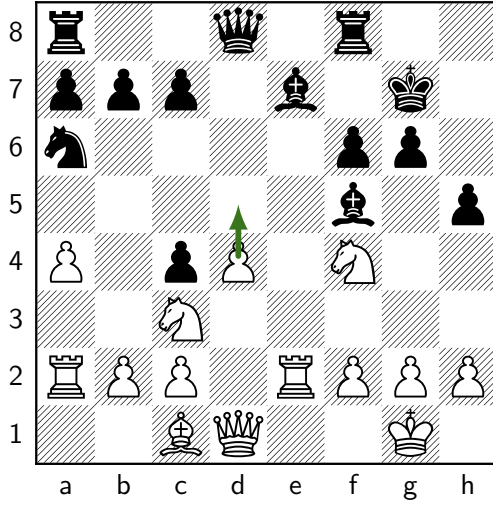


(c)

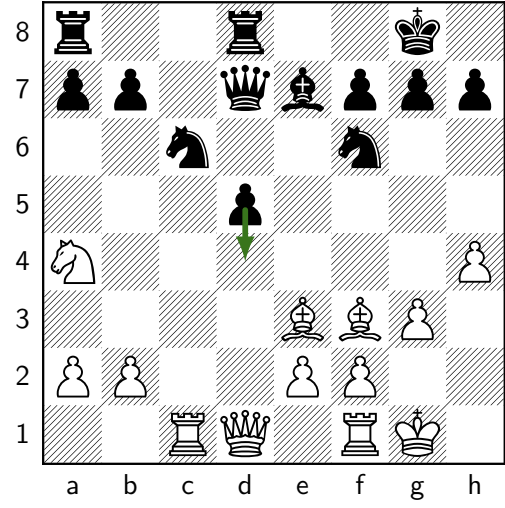


(d)

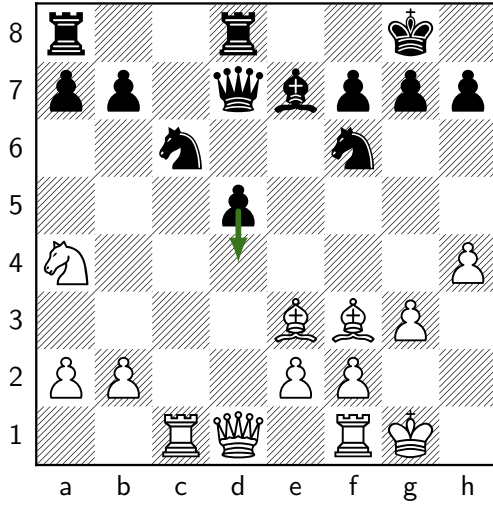
Figure 14: **Move 20, White: Nf4**. (a) Our position; Melanie plays Nf4 (h3f4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0001#3259 (ply 27) [structural distance $d=63$, QAP obj=65]: the master played Ne6; under μ our Nh3 plays the role of that game’s Nc4; cross-type roles: our pc7 as their Rf8, our na6 as their Pb7; transferred move h3f4 (**endorses this move**). (c) Retrieved from ccr1_top_p0001#3260 (ply 27) [structural distance $d=63$, QAP obj=65]: the master played Ne6; under μ our Nh3 plays the role of that game’s Nc4; cross-type roles: our pc7 as their Rf8, our na6 as their Pb7; transferred move h3f4 (**endorses this move**). (d) Retrieved from ccr1_top_p0002#2832 (ply 27) [structural distance $d=76$, QAP obj=50]: the master played Rfe8; under μ our Nh3 plays the role of that game’s Rf1 (a **cross-type** correspondence); cross-type roles: our Bc1 as their Ph7, our be7 as their Nf6; transferred move h3f4 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Nf4 (summed QAP objective 180, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



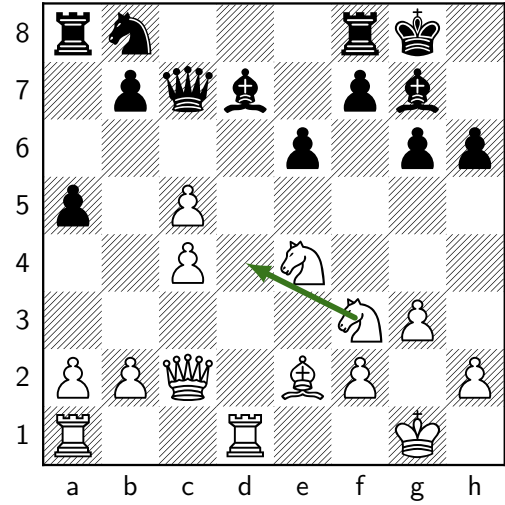
(a)



(b)

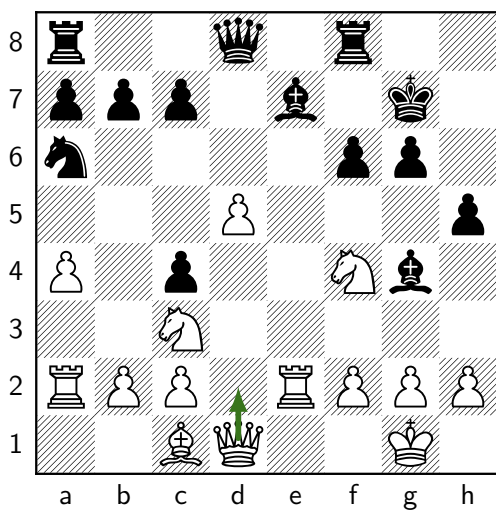


(c)

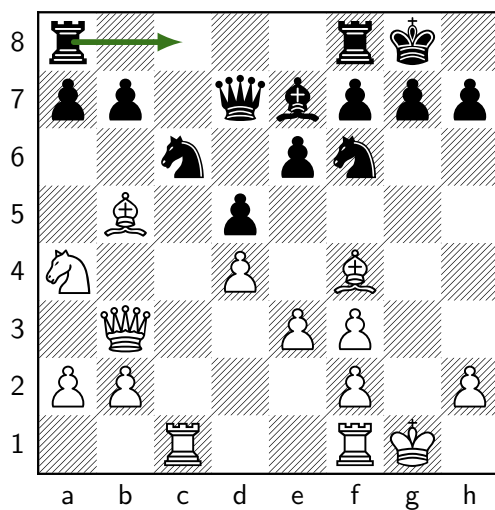


(d)

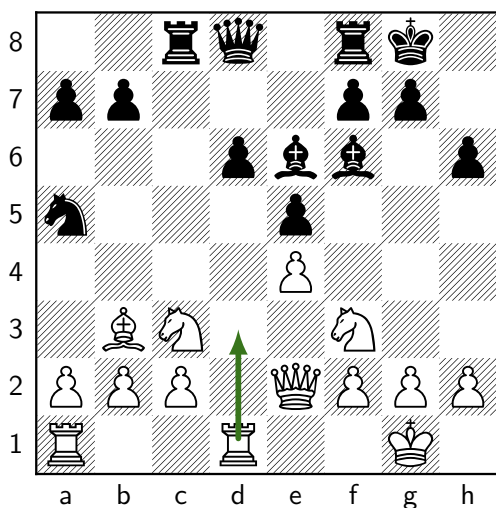
Figure 15: **Move 21, White: d5.** (a) Our position; Melanie plays d5 (d4d5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#1300` (ply 29) [structural distance $d=84$, QAP obj=73]: the master played d4; under μ our $Pd4$ plays the role of that game’s $Pd4$; cross-type roles: our $Bc1$ as their $Ra8$, our $Nc3$ as their $Rd8$; transferred move d4d5 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#2123` (ply 29) [structural distance $d=84$, QAP obj=73]: the master played d4; under μ our $Pd4$ plays the role of that game’s $Pd4$; cross-type roles: our $Bc1$ as their $Ra8$, our $Nc3$ as their $Rd8$; transferred move d4d5 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#3443` (ply 30) [structural distance $d=87$, QAP obj=66]: the master played Nd4; under μ our $Pb2$ plays the role of that game’s $Nf3$ (a **cross-type** correspondence); cross-type roles: our $Qd1$ as their $Ra1$, our $Bc1$ as their $Pa2$; transferred move b2b3. Synthesis: 2 retrieved analogs transferred to d5 (summed QAP objective 146, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



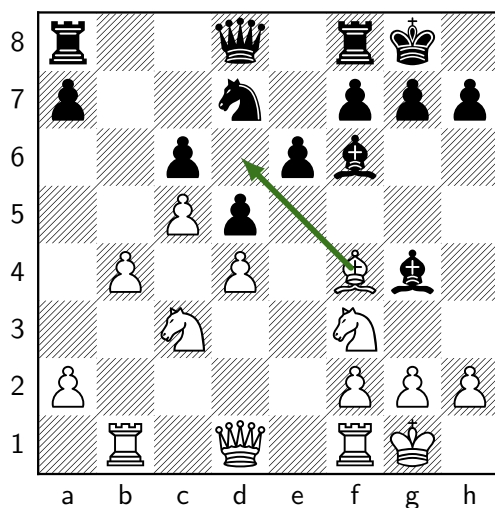
(a)



(b)

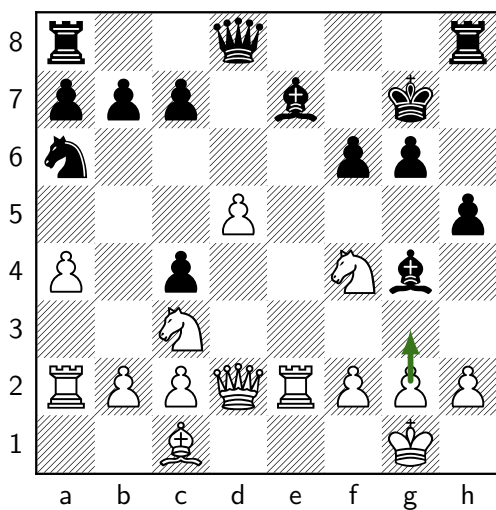


(c)

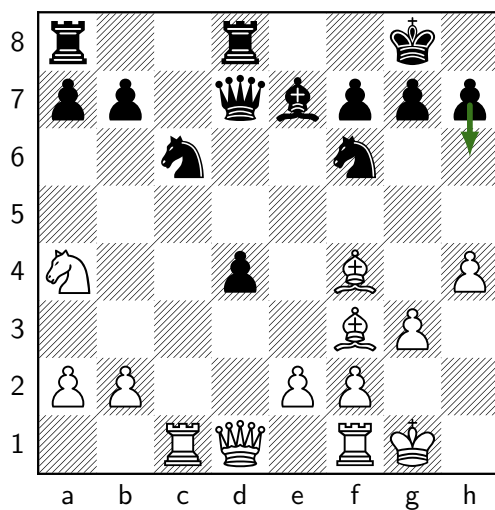


(d)

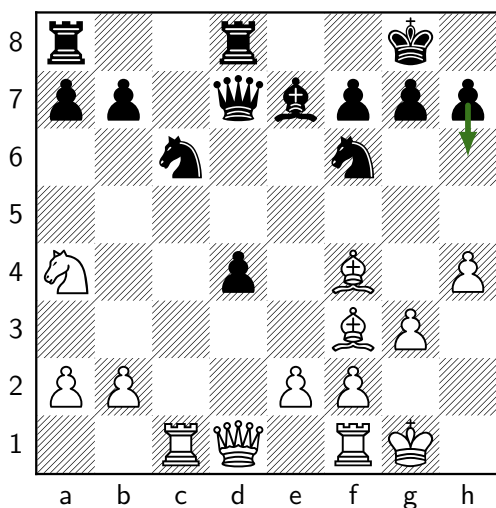
Figure 16: **Move 22, White: Qd2**. (a) Our position; Melanie plays Qd2 (d1d2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2375` (ply 27) [structural distance $d=118$, QAP obj=60]: the master played Rac8; under μ our Qd1 plays the role of that game’s Ra1 (a **cross-type** correspondence); cross-type roles: our Nf4 as their Rf8, our Nc3 as their Qd7; transferred move d1d2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3421` (ply 28) [structural distance $d=92$, QAP obj=26]: the master played Rd3; under μ our Qd1 plays the role of that game’s Rd1 (a **cross-type** correspondence); cross-type roles: our Bc1 as their Ra1, our Nc3 as their Pb2; transferred move d1d2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3200` (ply 26) [structural distance $d=112$, QAP obj=40]: the master played Bd6; under μ our Pd5 plays the role of that game’s Bf4 (a **cross-type** correspondence); cross-type roles: our Bc1 as their Rb1, our Pf2 as their Rf1; transferred move d5d6. Synthesis: 2 retrieved analogs transferred to Qd2 (summed QAP objective 85, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



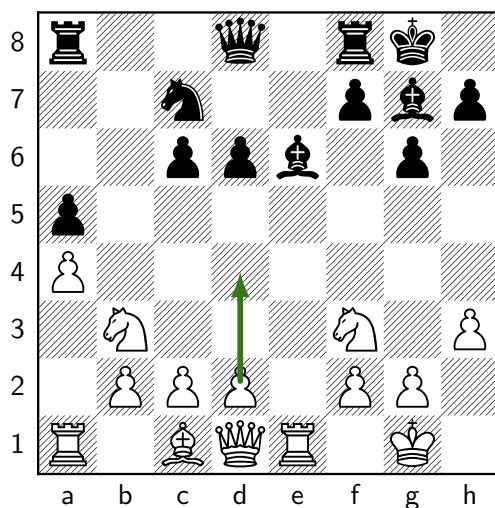
(a)



(b)

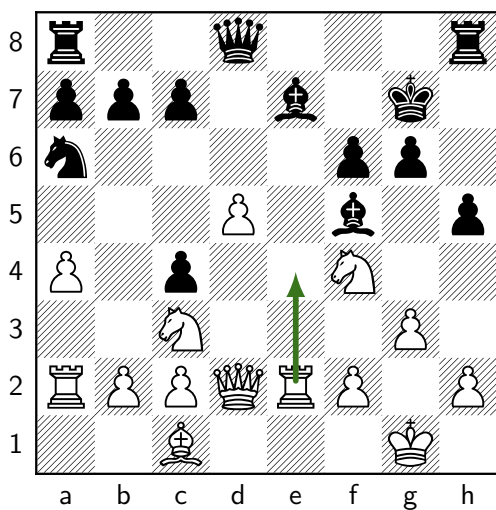


(c)

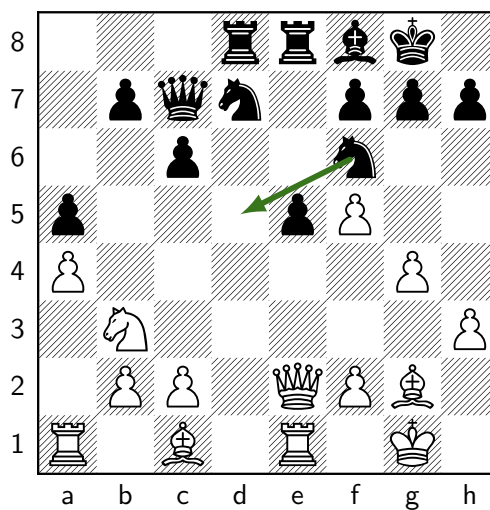


(d)

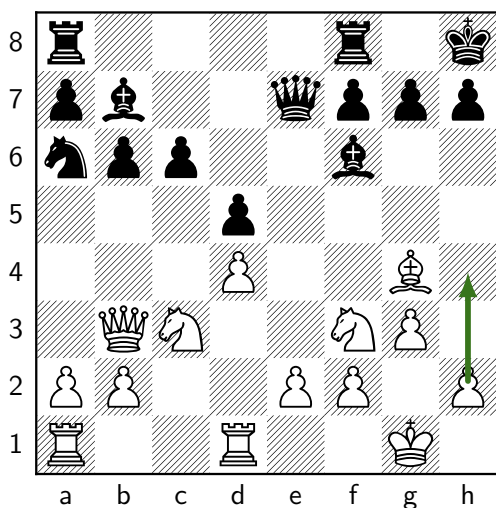
Figure 17: **Move 23, White: g3.** (a) Our position; Melanie plays g3 (g2g3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1300` (ply 31) [structural distance $d=98$, QAP obj=46]: the master played h6; under μ our $Pg2$ plays the role of that game’s $Ph2$; cross-type roles: our $Bc1$ as their $Ra8$, our $Qd2$ as their $Rd8$; transferred move g2g3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2123` (ply 31) [structural distance $d=98$, QAP obj=46]: the master played h6; under μ our $Pg2$ plays the role of that game’s $Ph2$; cross-type roles: our $Bc1$ as their $Ra8$, our $Qd2$ as their $Rd8$; transferred move g2g3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#2822` (ply 28) [structural distance $d=107$, QAP obj=44]: the master played d4; under μ our $Nc3$ plays the role of that game’s $Pd2$ (a **cross-type** correspondence); cross-type roles: our $pc4$ as their $Ra1$, our $pb7$ as their $Bc1$; transferred move c3e4. Synthesis: 2 retrieved analogs transferred to g3 (summed QAP objective 92, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



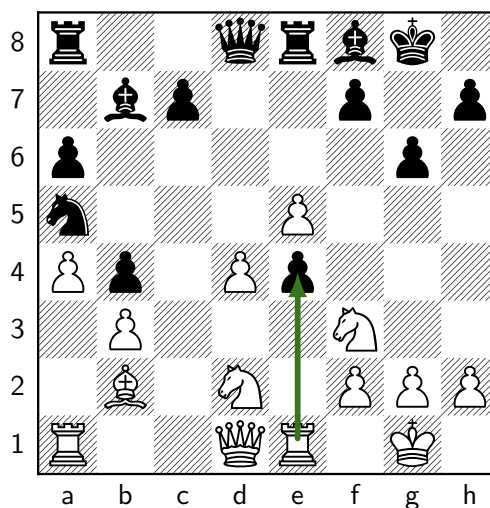
(a)



(b)

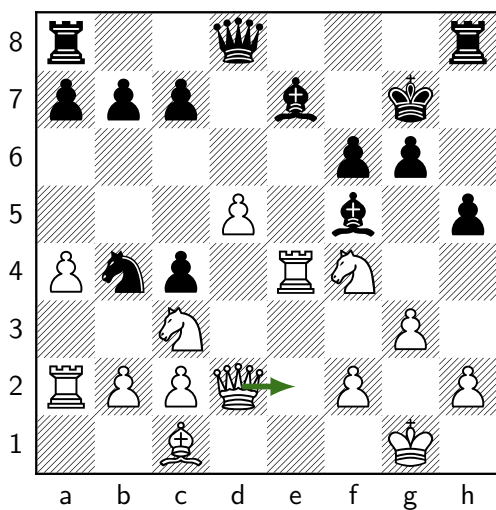


(c)

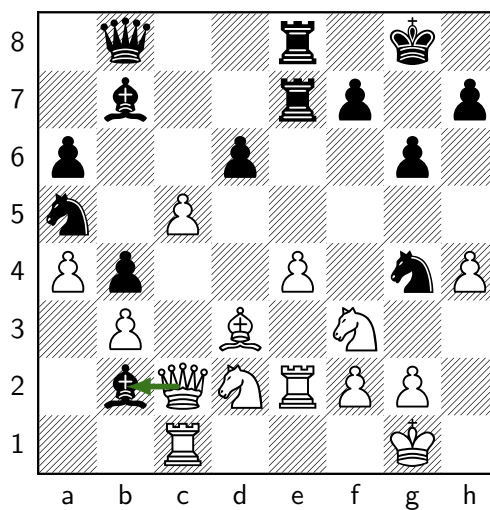


(d)

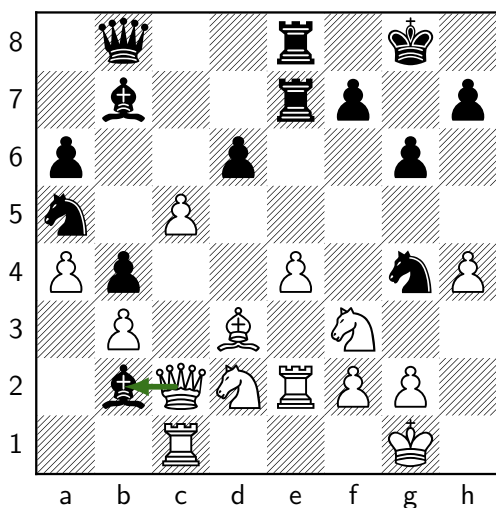
Figure 18: **Move 24, White: Re4.** (a) Our position; Melanie plays Re4 (e2e4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#1008` (ply 35) [structural distance $d=83$, QAP obj=75]: the master played Nd5; under μ our *Re2* plays the role of that game’s *Nf3* (a **cross-type** correspondence); cross-type roles: our *pc4* as their *Rd8*, our *na6* as their *Re8*; transferred move e2e4 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#391` (ply 30) [structural distance $d=90$, QAP obj=71]: the master played h4; under μ our *Re2* plays the role of that game’s *Ph2* (a **cross-type** correspondence); cross-type roles: our *Qd2* as their *Ra1*, our *pf6* as their *Rd1*; transferred move e2e4 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#3035` (ply 36) [structural distance $d=60$, QAP obj=25]: the master played Rxe4; under μ our *Re2* plays the role of that game’s *Re1*; cross-type roles: our *Pb2* as their *Ra1*, our *Nf4* as their *Bb2*; transferred move e2e4 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Re4 (summed QAP objective 171, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



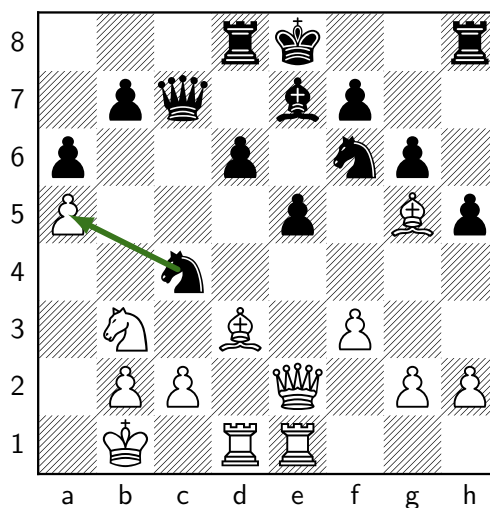
(a)



(b)

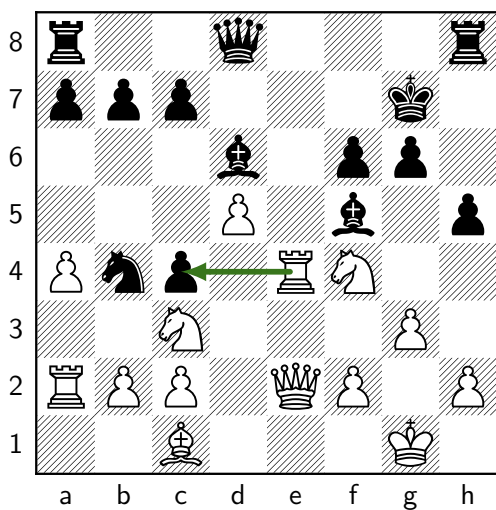


(c)

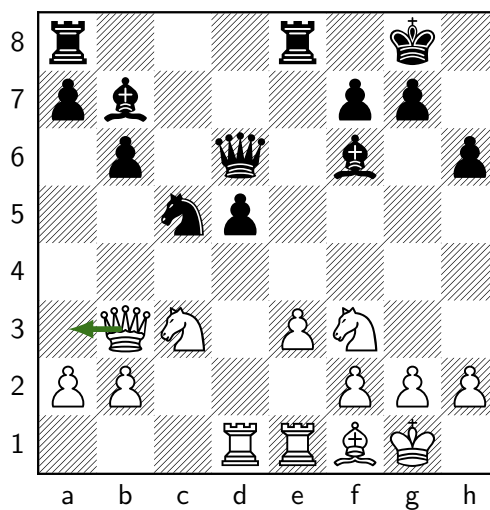


(d)

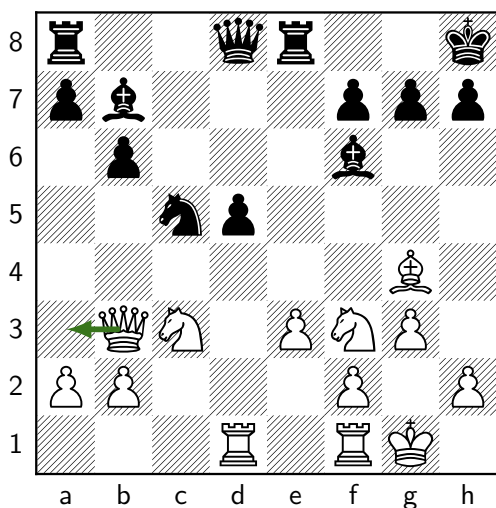
Figure 19: **Move 25, White: Qe2.** (a) Our position; Melanie plays Qe2 (d2e2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3727` (ply 54) [structural distance $d=128$, QAP obj=52]: the master played Qxb2; under μ our Nc3 plays the role of that game’s Qc2 (a **cross-type** correspondence); cross-type roles: our Qd2 as their Nd2, our Nf4 as their Re2; transferred move c3b1. (c) Retrieved from `ccr1_top_p0000#1395` (ply 58) [structural distance $d=128$, QAP obj=52]: the master played Qxb2; under μ our Nc3 plays the role of that game’s Qc2 (a **cross-type** correspondence); cross-type roles: our Qd2 as their Nd2, our Nf4 as their Re2; transferred move c3b1. (d) Retrieved from `ccr1_top_p0000#2926` (ply 37) [structural distance $d=126$, QAP obj=48]: the master played Nxa5; under μ our Re4 plays the role of that game’s Nc5 (a **cross-type** correspondence); cross-type roles: our Ph2 as their Rd8, our Nf4 as their Rh8; transferred move e4e1. Synthesis: no retrieved analog transferred to Qe2 – the search selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



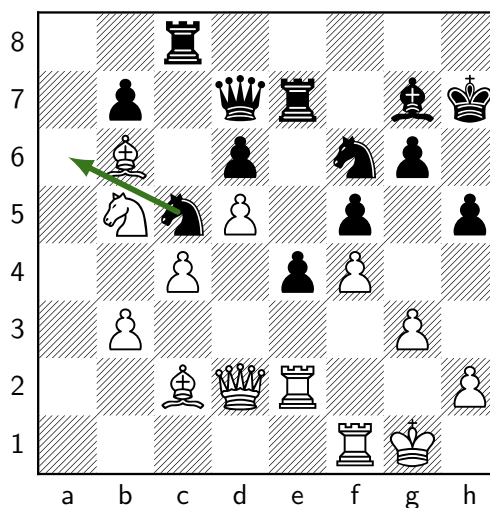
(a)



(b)



(c)



(d)

Figure 20: **Move 26, White: Rxc4.** (a) Our position; Melanie plays Rxc4 (e4c4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1558` (ply 34) [structural distance $d=130$, QAP obj=69]: the master played Qa3; under μ our *Re4* plays the role of that game’s *Qb3* (a **cross-type** correspondence); cross-type roles: our *Nf4* as their *Re1*, our *Nc3* as their *Bf1*; transferred move e4c4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3066` (ply 36) [structural distance $d=130$, QAP obj=68]: the master played Qa3; under μ our *Re4* plays the role of that game’s *Qb3* (a **cross-type** correspondence); cross-type roles: our *Nf4* as their *Rf1*, our *Nc3* as their *Pa2*; transferred move e4c4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#3617` (ply 67) [structural distance $d=131$, QAP obj=52]: the master played Na6; under μ our *Re4* plays the role of that game’s *Nc4* (a **cross-type** correspondence); cross-type roles: our *pc7* as their *Rc8*, our *bd6* as their *Qd7*; transferred move e4c4 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Rxc4 (summed QAP objective 251, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.

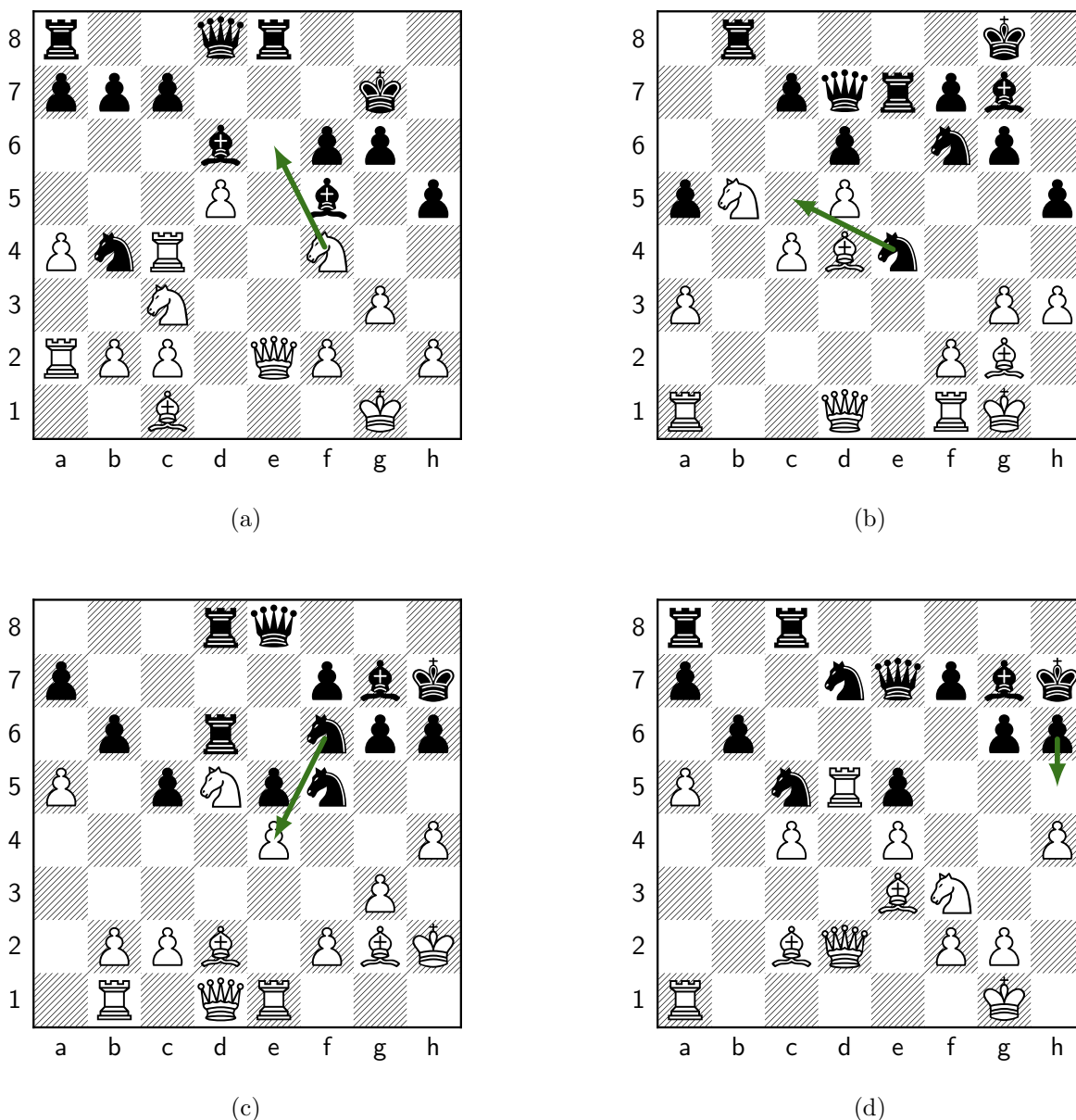
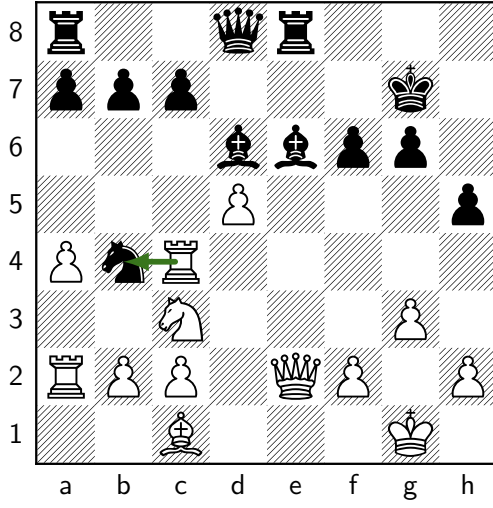
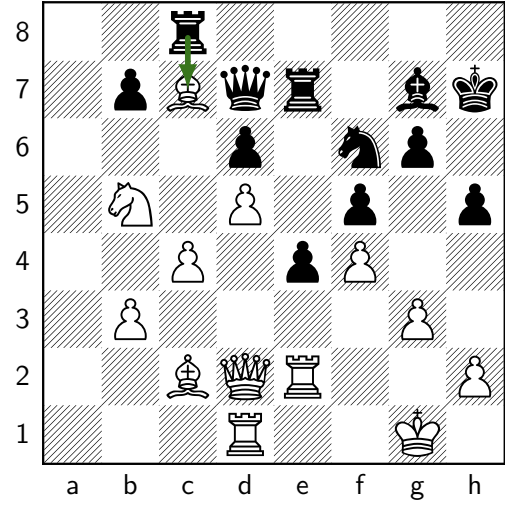


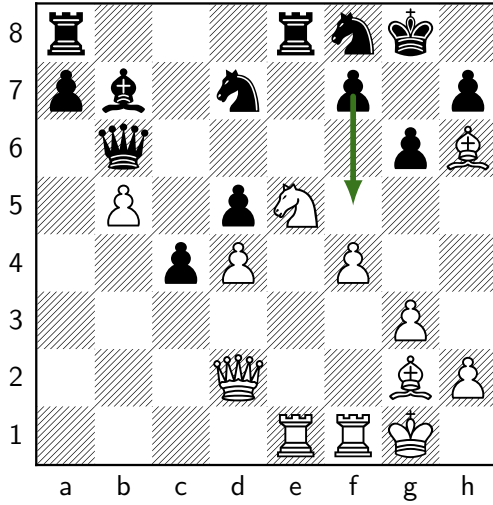
Figure 21: **Move 27, White: Ne6+.** (a) Our position; Melanie plays Ne6+ (f4e6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0002#2191 (ply 43) [structural distance $d=141$, QAP obj=36]: the master played Nc5; under μ our Nf4 plays the role of that game’s Ne5; cross-type roles: our Nc3 as their Pc7, our Rc4 as their Qd7; transferred move f4e6 (**endorses this move**). (c) Retrieved from ccr1_top_p0000#1425 (ply 47) [structural distance $d=124$, QAP obj=54]: the master played Nxe4; under μ our Rc4 plays the role of that game’s Nf3 (a **cross-type** correspondence); cross-type roles: our Pa4 as their Rd8, our Nc3 as their Qe8; transferred move c4c5. (d) Retrieved from ccr1_top_p0003#2294 (ply 57) [structural distance $d=134$, QAP obj=49]: the master played h5; under μ our Ph2 plays the role of that game’s Ph3; cross-type roles: our Bc1 as their Ra8, our qd8 as their Rc8; transferred move h2h4. Synthesis: 1 retrieved analog transferred to Ne6+ (summed QAP objective 36, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



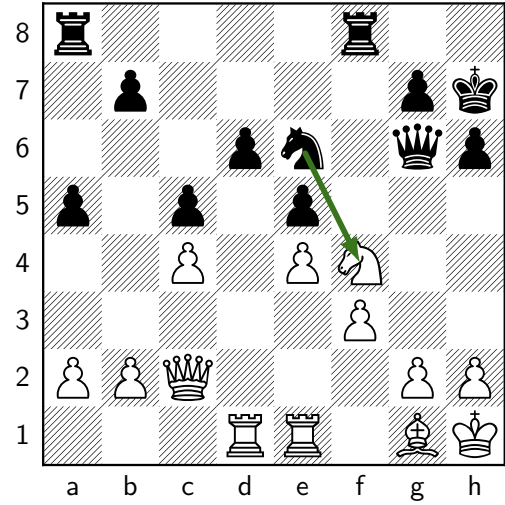
(a)



(b)

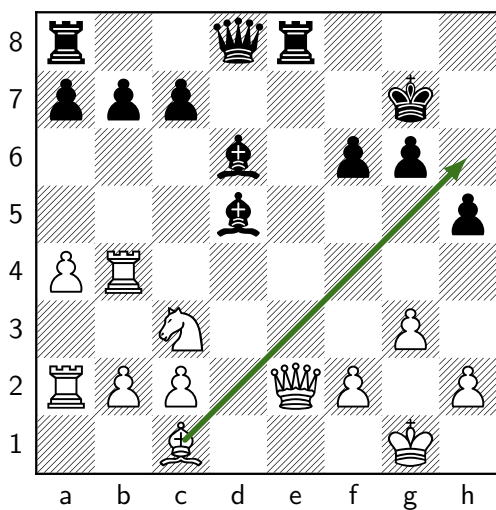


(c)

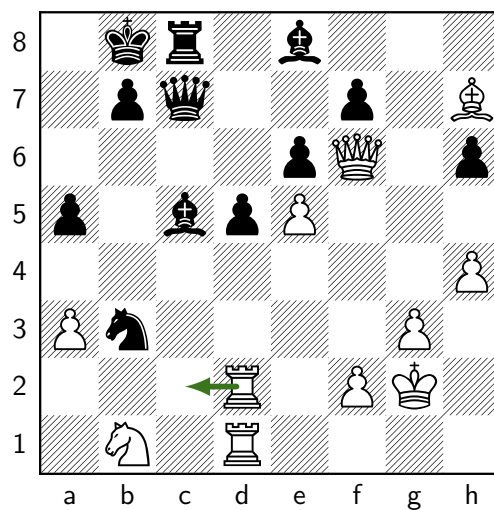


(d)

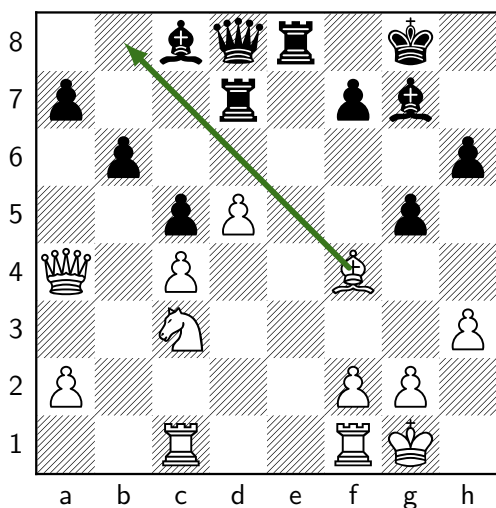
Figure 22: **Move 28, White: Rxb4.** (a) Our position; Melanie plays Rxb4 (c4b4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3617` (ply 71) [structural distance $d=104$, QAP obj=68]: the master played Rxc7; under μ our *Rc4* plays the role of that game’s *Rc1*; cross-type roles: our *nb4* as their *bc7*, our *Rc4* as their *Qd7*; transferred move c4b4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#567` (ply 65) [structural distance $d=112$, QAP obj=60]: the master played f5; under μ our *Ph2* plays the role of that game’s *Pf2*; cross-type roles: our *Pc2* as their *Ra8*, our *Qe2* as their *Re8*; transferred move h2h4. (d) Retrieved from `ccr1_top_p0002#1850` (ply 51) [structural distance $d=113$, QAP obj=59]: the master played Nxf4; under μ our *Ra2* plays the role of that game’s *Ne3* (a **cross-type** correspondence); cross-type roles: our *Qe2* as their *Ra8*, our *Nc3* as their *Pg7*; transferred move a2a3. Synthesis: 1 retrieved analog transferred to Rxb4 (summed QAP objective 68, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



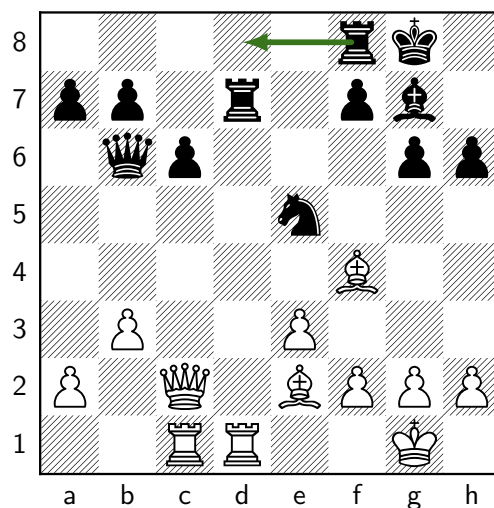
(a)



(b)

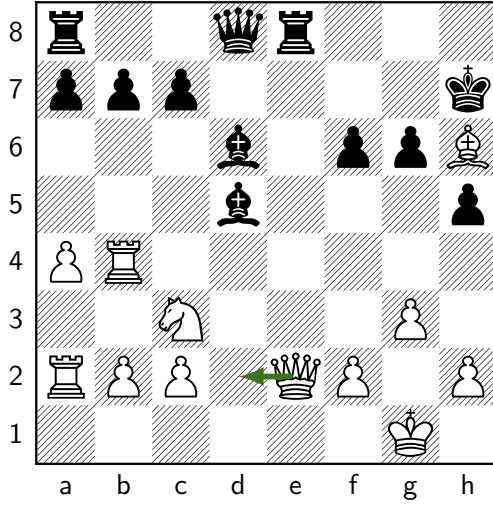


(c)

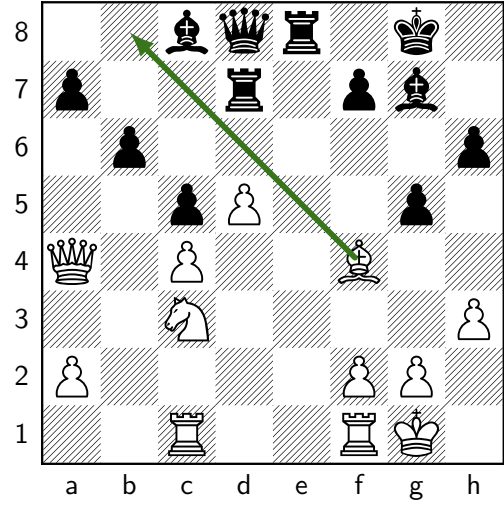


(d)

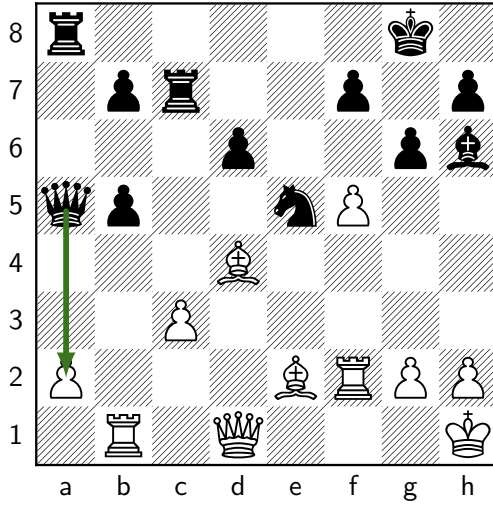
Figure 23: **Move 29, White: Bh6+.** (a) Our position; Melanie plays Bh6+ (c1h6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2955` (ply 60) [structural distance $d=106$, QAP obj=32]: the master played Rc2; under μ our Rb4 plays the role of that game’s Rd2; cross-type roles: our ph5 as their Nb1, our Pf2 as their Rd1; transferred move b4h4. (c) Retrieved from `ccr1_top_p0003#2780` (ply 42) [structural distance $d=103$, QAP obj=32]: the master played Bb8; under μ our Qe2 plays the role of that game’s Bf4 (a **cross-type** correspondence); cross-type roles: our pc7 as their Rc1, our Pf2 as their Rf1; transferred move e2e8. (d) Retrieved from `ccr1_top_p0003#2724` (ply 41) [structural distance $d=89$, QAP obj=31]: the master played Rfd8; under μ our Ph2 plays the role of that game’s Rf1 (a **cross-type** correspondence); cross-type roles: our Qe2 as their Rd7, our Pb2 as their Bg7; transferred move h2h4. Synthesis: no retrieved analog transferred to Bh6+ – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



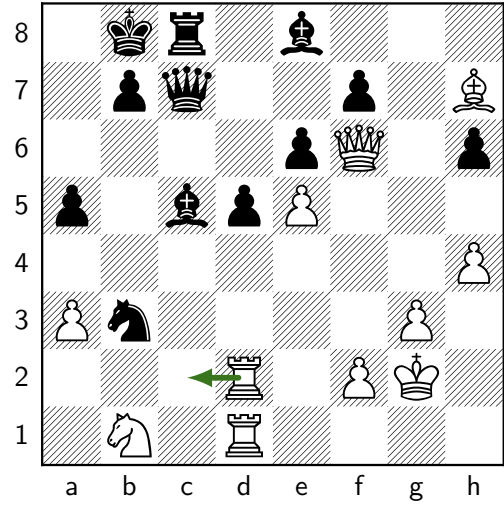
(a)



(b)

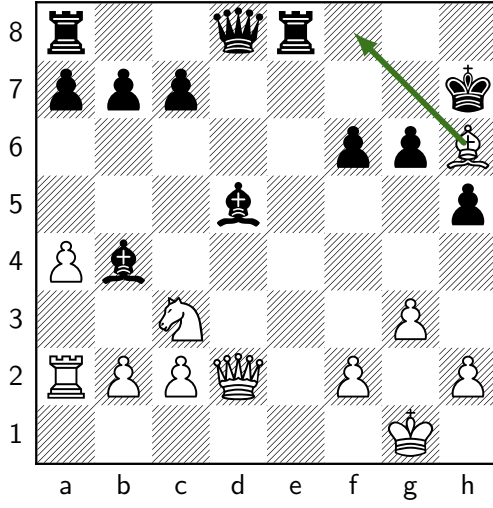


(c)

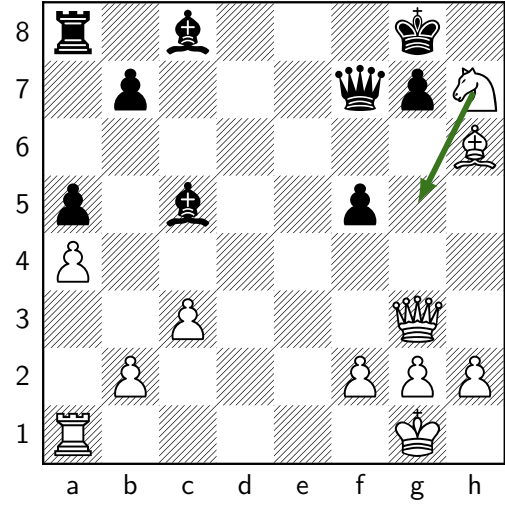


(d)

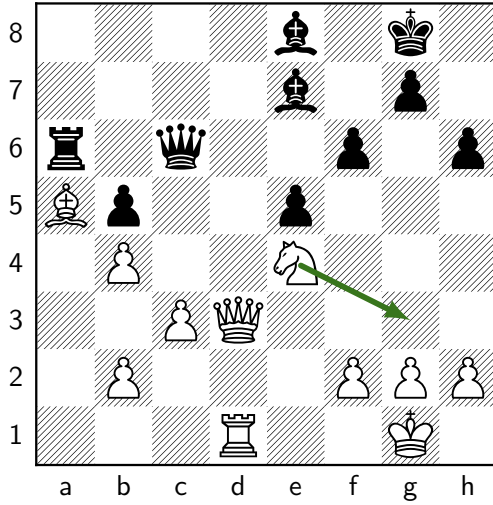
Figure 24: **Move 30, White: Qd2.** (a) Our position; Melanie plays Qd2 (e2d2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2780` (ply 42) [structural distance $d=105$, QAP obj=32]: the master played Bb8; under μ our Qe2 plays the role of that game’s Bf4 (a **cross-type** correspondence); cross-type roles: our bd5 as their Nc3, our bd6 as their Qa4; transferred move e2e8. (c) Retrieved from `ccr1_top_p0001#1319` (ply 45) [structural distance $d=86$, QAP obj=30]: the master played Qxa2; under μ our Bh6 plays the role of that game’s Qa4 (a **cross-type** correspondence); cross-type roles: our Pc2 as their Ra8, our Ph2 as their Rc7; transferred move h6g5. (d) Retrieved from `ccr1_top_p0000#2955` (ply 60) [structural distance $d=106$, QAP obj=25]: the master played Rc2; under μ our Rb4 plays the role of that game’s Rd2; cross-type roles: our Pb2 as their Nb1, our Pg3 as their Rd1; transferred move b4h4. Synthesis: no retrieved analog transferred to Qd2 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



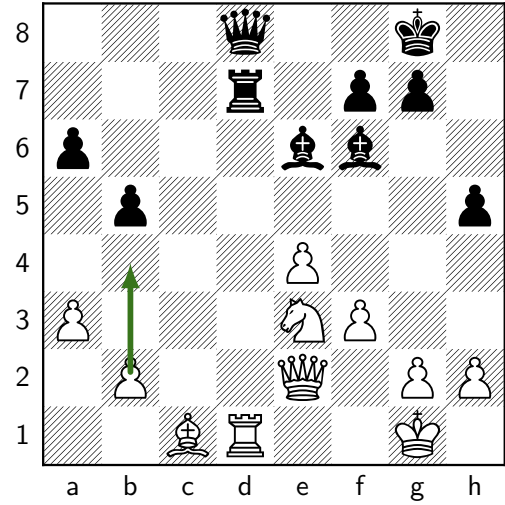
(a)



(b)

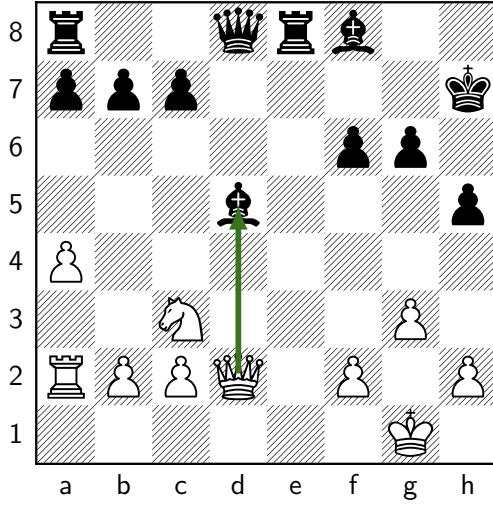


(c)

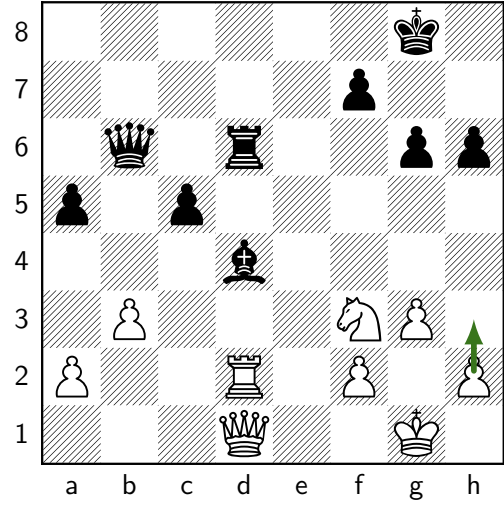


(d)

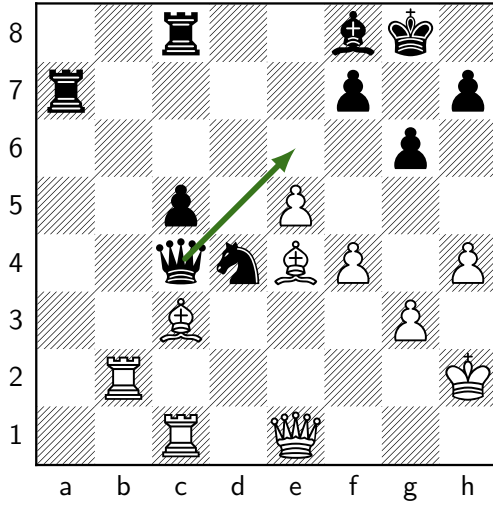
Figure 25: **Move 31, White: Bf8**. (a) Our position; Melanie plays Bf8 (h6f8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3564` (ply 46) [structural distance $d=92$, QAP obj=50]: the master played Ng5; under μ our Bh6 plays the role of that game’s Nh7 (a **cross-type** correspondence); cross-type roles: our pg6 as their Ra1, our re8 as their Pg2; transferred move h6g5. (c) Retrieved from `ccr1_top_p0002#3660` (ply 52) [structural distance $d=88$, QAP obj=31]: the master played Ng3; under μ our Ra2 plays the role of that game’s Ne4 (a **cross-type** correspondence); cross-type roles: our Pg3 as their Rd1, our Bh6 as their Pc3; transferred move a2a3. (d) Retrieved from `ccr1_top_p0000#1266` (ply 58) [structural distance $d=91$, QAP obj=17]: the master played b4; under μ our Nc3 plays the role of that game’s Pb2 (a **cross-type** correspondence); cross-type roles: our Pb2 as their Bc1, our Qd2 as their Rd1; transferred move c3b1. Synthesis: no retrieved analog transferred to Bf8 – the **search selector** chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



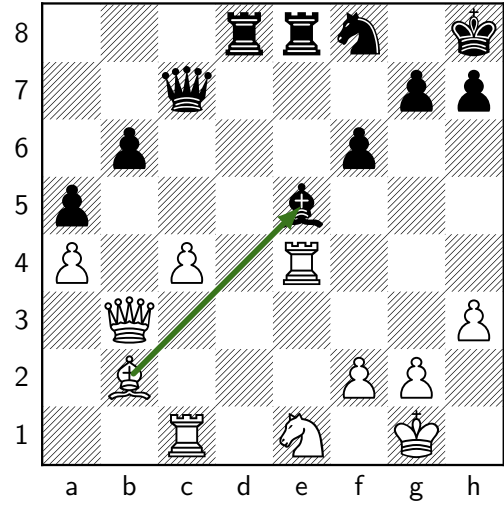
(a)



(b)

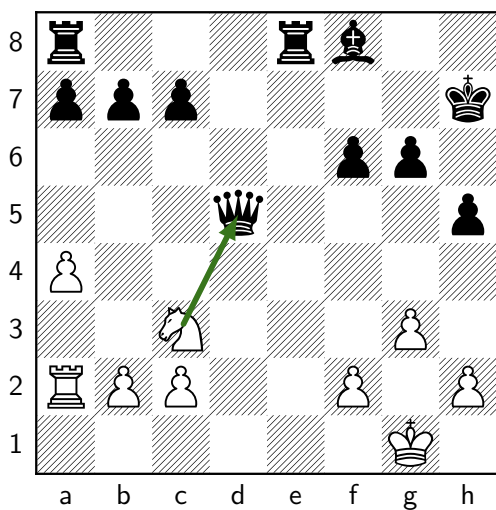


(c)

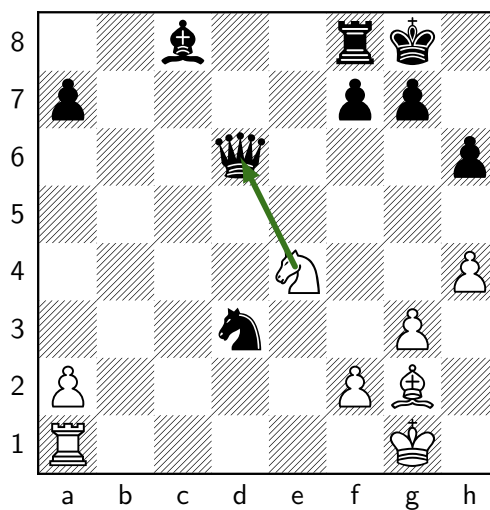


(d)

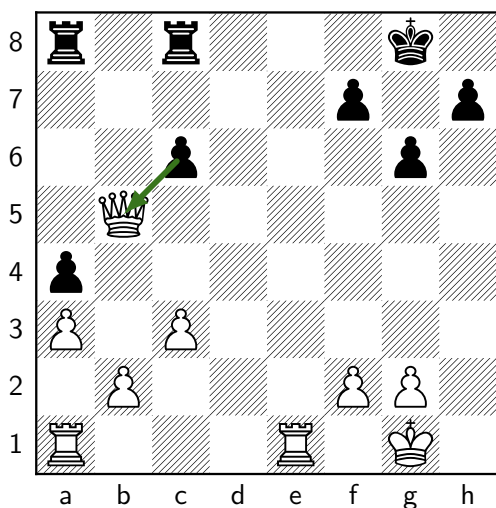
Figure 26: **Move 32, White: Qxd5.** (a) Our position; Melanie plays Qxd5 (d2d5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#500` (ply 70) [structural distance $d=81$, QAP obj=75]: the master played h3; under μ our *Pa4* plays the role of that game’s *Ph2*; cross-type roles: our *Pc2* as their *Qd1*, our *Qd2* as their *Rd2*; transferred move a4a5. (c) Retrieved from `ccr1_top_p0002#920` (ply 77) [structural distance $d=84$, QAP obj=51]: the master played Qe6; under μ our *Qd2* plays the role of that game’s *Qc5*; cross-type roles: our *Pf2* as their *Rc8*, our *Nc3* as their *Bf8*; transferred move d2f4. (d) Retrieved from `ccr1_top_p0000#1923` (ply 52) [structural distance $d=92$, QAP obj=43]: the master played Bxe5; under μ our *Ra2* plays the role of that game’s *Bb2* (a **cross-type** correspondence); cross-type roles: our *Nc3* as their *Rc1*, our *Pf2* as their *Ne1*; transferred move a2a3. Synthesis: no retrieved analog transferred to Qxd5 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



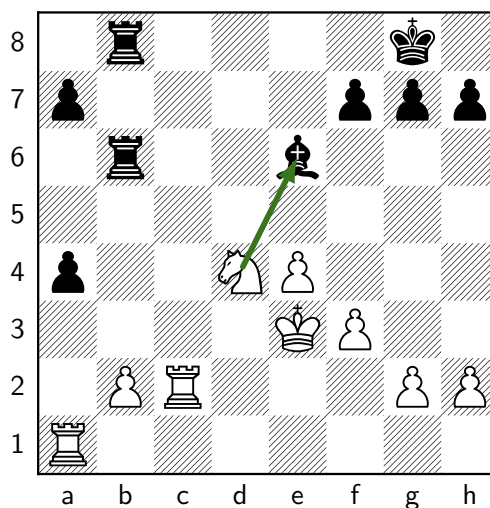
(a)



(b)

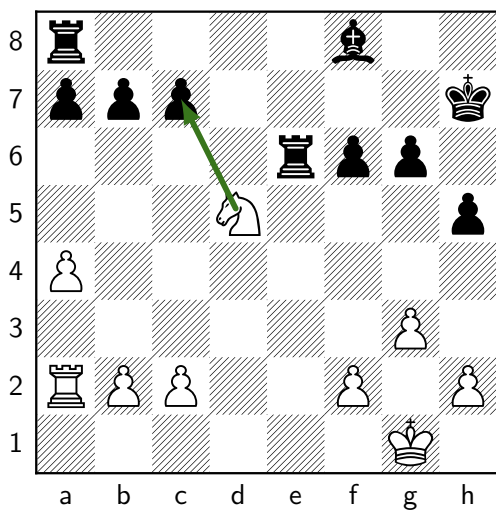


(c)

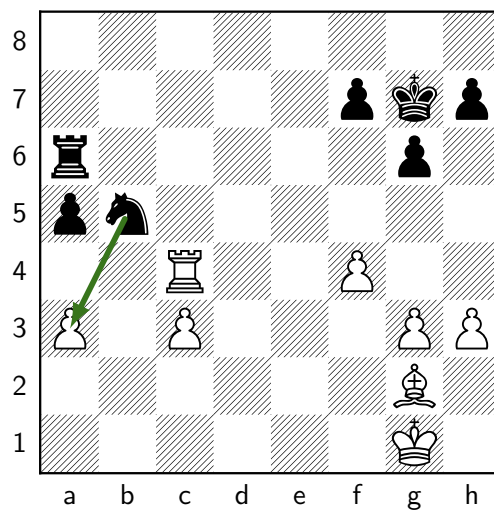


(d)

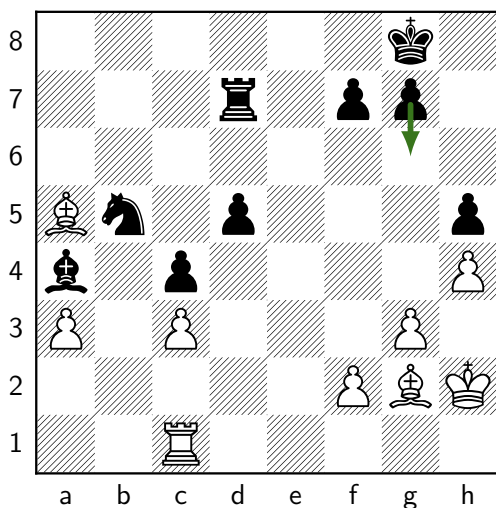
Figure 27: **Move 33, White: Nxd5.** (a) Our position; Melanie plays Nxd5 (c3d5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1937` (ply 50) [structural distance $d=71$, QAP obj=23]: the master played Nxd6; under μ our Nc3 plays the role of that game's Ne4; cross-type roles: our bf8 as their Ra1, our Pf2 as their Bg2; transferred move c3d5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#2859` (ply 53) [structural distance $d=77$, QAP obj=13]: the master played cxb5; under μ our Nc3 plays the role of that game's Pc3 (a **cross-type** correspondence); cross-type roles: our Pf2 as their Ra8, our Pc2 as their Rc8; transferred move c3d5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#236` (ply 54) [structural distance $d=73$, QAP obj=9]: the master played Nxe6; under μ our Nc3 plays the role of that game's Nd4; cross-type roles: our pa7 as their Ra1, our bf8 as their Pf3; transferred move c3d5 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Nxd5 (summed QAP objective 45, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



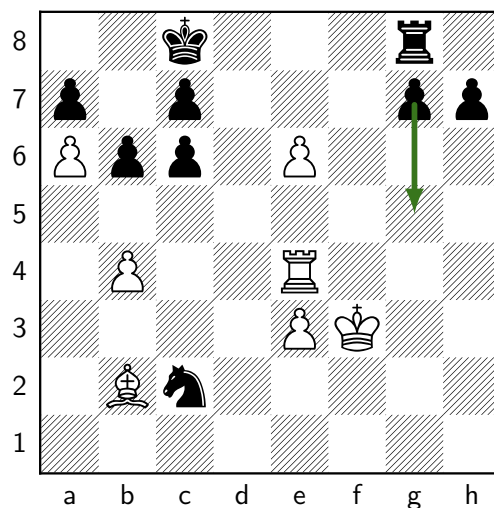
(a)



(b)

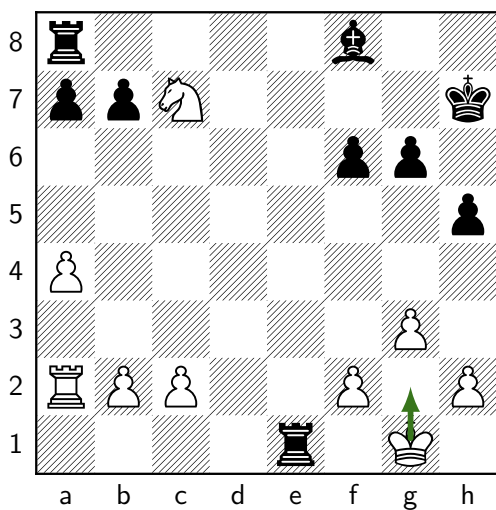


(c)

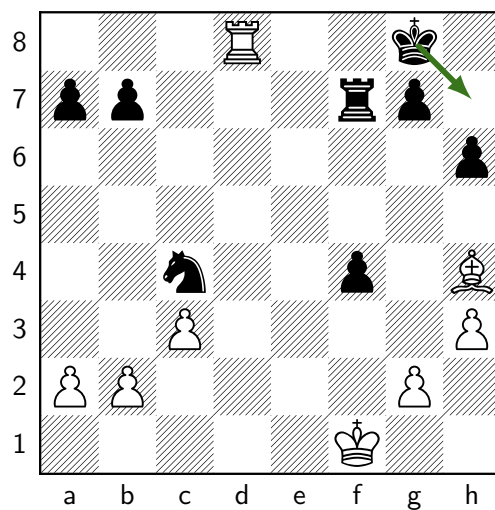


(d)

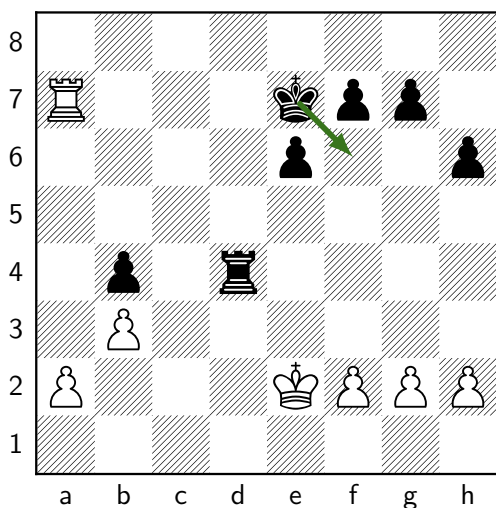
Figure 28: **Move 34, White: Nxc7**. (a) Our position; Melanie plays Nxc7 (d5c7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top-p0001#1106` (ply 77) [structural distance $d=37$, QAP obj=47]: the master played Nxa3; under μ our Nd5 plays the role of that game’s Nb4; cross-type roles: our ra8 as their bg2; transferred move d5c7 (**endorses this move**). (c) Retrieved from `ccr1_top-p0002#364` (ply 53) [structural distance $d=37$, QAP obj=53]: the master played g6; under μ our Ph2 plays the role of that game’s Pg2; cross-type roles: our re6 as their ba5, our Nd5 as their bg2; transferred move h2h4. (d) Retrieved from `ccr1_top-p0003#2073` (ply 61) [structural distance $d=43$, QAP obj=45]: the master played g5; under μ our Pa4 plays the role of that game’s Pg2; transferred move a4a5. Synthesis: 1 retrieved analog transferred to Nxc7 (summed QAP objective 47, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



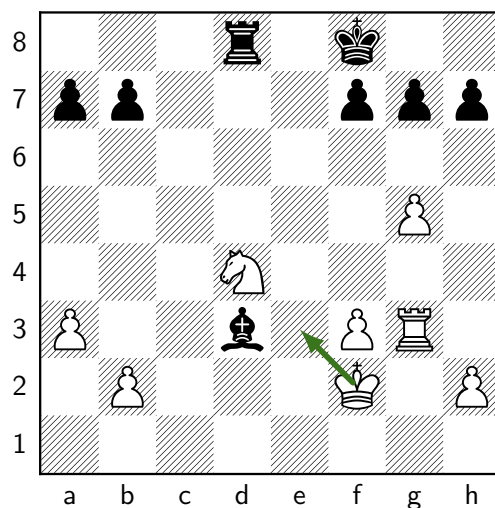
(a)



(b)

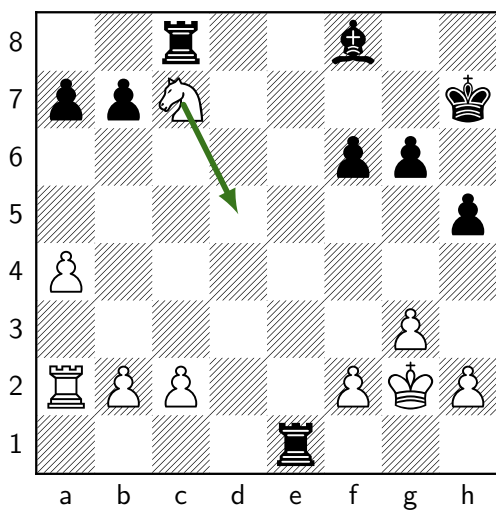


(c)

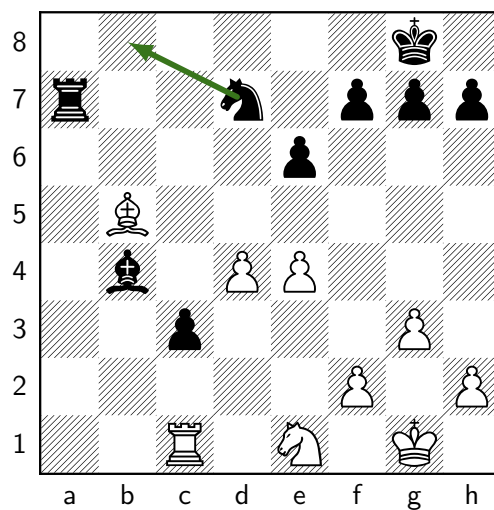


(d)

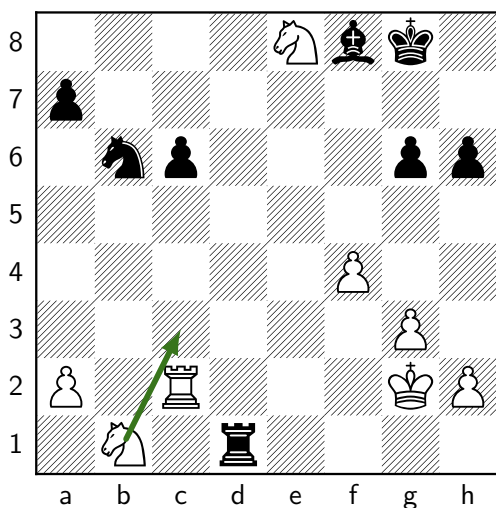
Figure 29: **Move 35, White: Kg2.** (a) Our position; Melanie plays Kg2 (**g1g2**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#3882` (ply 61) [structural distance $d=35$, QAP obj=51]: the master played Kh7; under μ our *Kg1* plays the role of that game’s *Kg1*; cross-type roles: our *Ra2* as their *Pa7*, our *Pf2* as their *Rf7*; transferred move **g1g2 (endorses this move)**. (c) Retrieved from `ccrl_top_p0000#309` (ply 57) [structural distance $d=35$, QAP obj=46]: the master played Kf6; under μ our *Kg1* plays the role of that game’s *Ke2*; cross-type roles: our *Ra2* as their *Pg7*, our *Nc7* as their *pb3*; transferred move **g1g2 (endorses this move)**. (d) Retrieved from `ccrl_top_p0002#1548` (ply 62) [structural distance $d=38$, QAP obj=3]: the master played Ke3; under μ our *Kg1* plays the role of that game’s *Kf2*; cross-type roles: our *bf8* as their *Pa3*, our *Pg3* as their *bd3*; transferred move **g1g2 (endorses this move)**. Synthesis: 3 retrieved analogs transferred to Kg2 (summed QAP objective 100, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



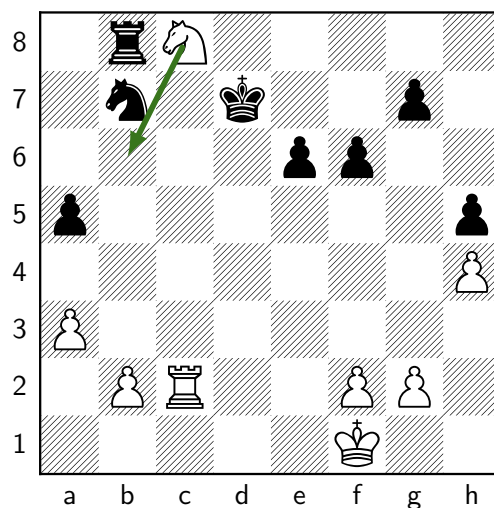
(a)



(b)

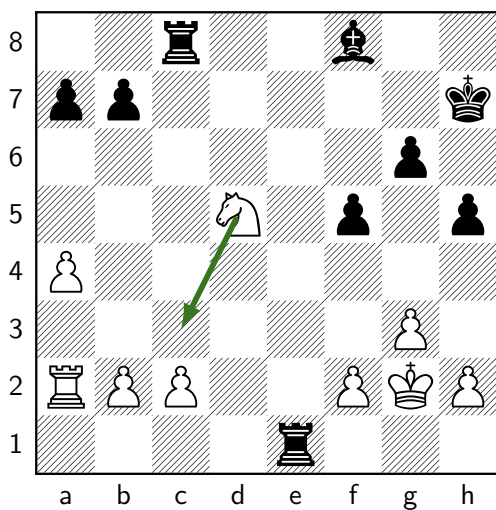


(c)

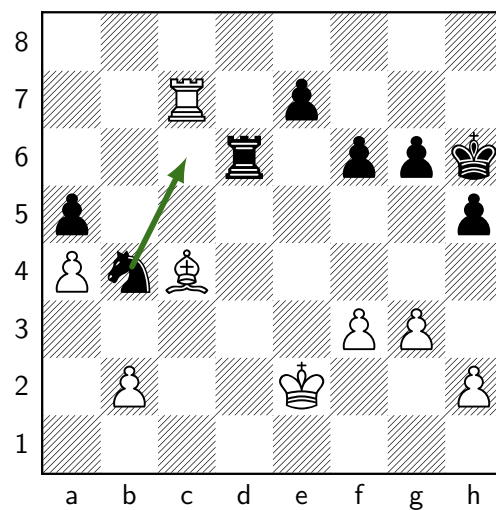


(d)

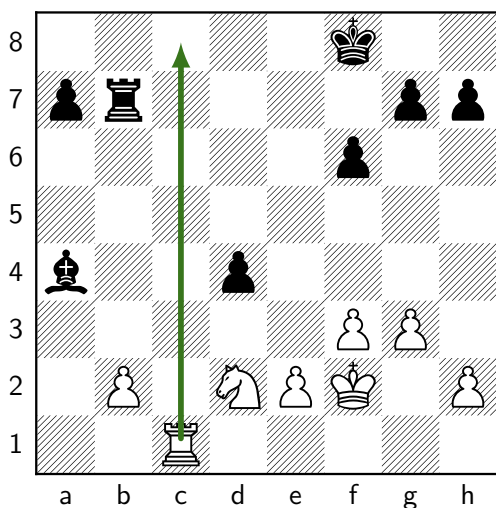
Figure 30: **Move 36, White: Nd5.** (a) Our position; Melanie plays Nd5 (c7d5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#177` (ply 59) [structural distance $d=38$, QAP obj=16]: the master played Nb8; under μ our Nc7 plays the role of that game’s Nd2; cross-type roles: our pf6 as their Ra7, our rc8 as their bb5; transferred move c7d5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3173` (ply 60) [structural distance $d=37$, QAP obj=12]: the master played Nc3; under μ our Nc7 plays the role of that game’s Nb1; cross-type roles: our Ra2 as their Pa2, our Pb2 as their Rc2; transferred move c7d5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#614` (ply 66) [structural distance $d=37$, QAP obj=22]: the master played Nb6+; under μ our Nc7 plays the role of that game’s Nc8; cross-type roles: our re1 as their Pa3, our bf8 as their nb7; transferred move c7a8. Synthesis: 2 retrieved analogs transferred to Nd5 (summed QAP objective 28, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



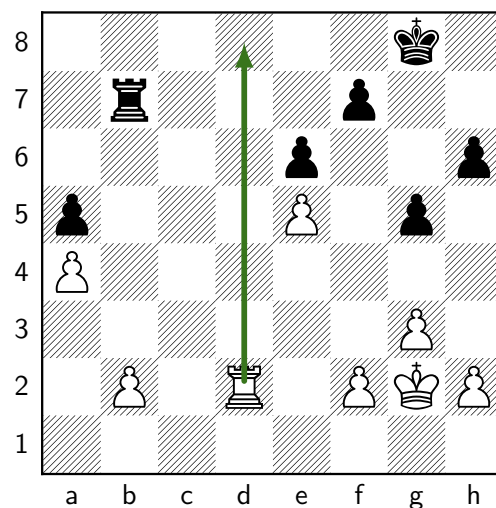
(a)



(b)

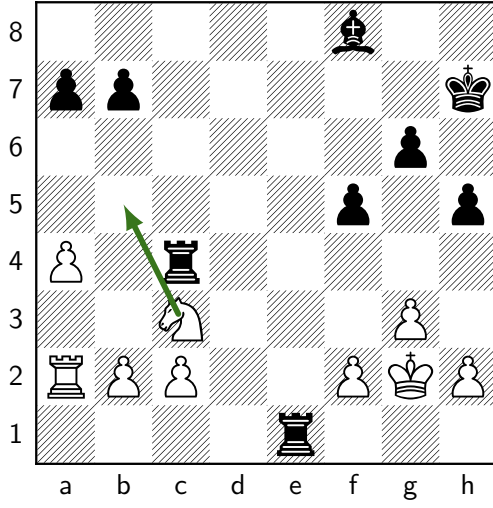


(c)

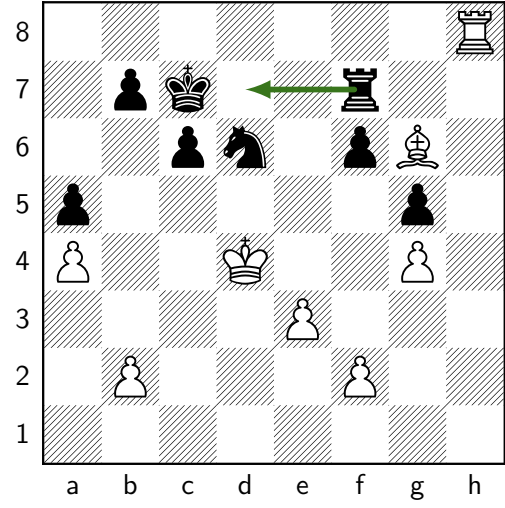


(d)

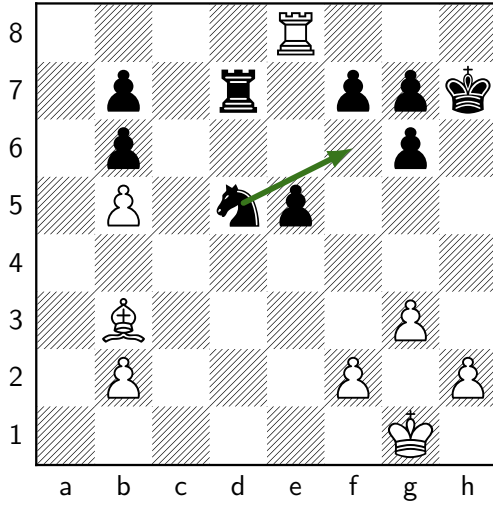
Figure 31: **Move 37, White: Nc3.** (a) Our position; Melanie plays Nc3 (d5c3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#2304` (ply 69) [structural distance $d=37$, QAP obj=10]: the master played Nc6; under μ our Nd5 plays the role of that game’s Nb5; cross-type roles: our Pb2 as their Rd6, our re1 as their Pf6; transferred move d5c3 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#2498` (ply 72) [structural distance $d=36$, QAP obj=31]: the master played Rc8+; under μ our Nd5 plays the role of that game’s Rc1 (a **cross-type** correspondence); cross-type roles: our Pb2 as their Nd2, our pa7 as their ba4; transferred move d5c7. (d) Retrieved from `ccrl_top_p0001#3314` (ply 68) [structural distance $d=25$, QAP obj=29]: the master played Rd8+; under μ our Nd5 plays the role of that game’s Rd2 (a **cross-type** correspondence); cross-type roles: our bf8 as their pa5, our re1 as their Pe5; transferred move d5c7. Synthesis: 1 retrieved analog transferred to Nc3 (summed QAP objective 10, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



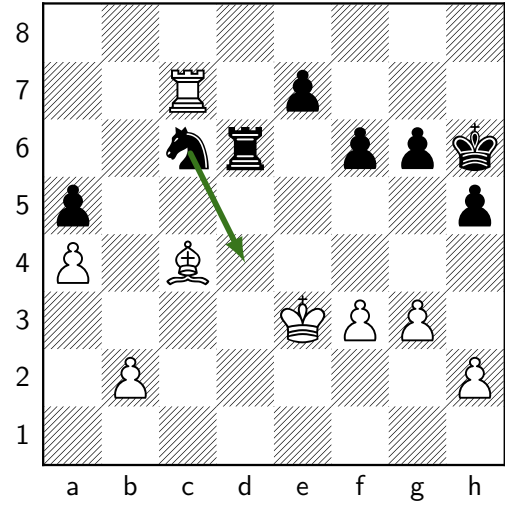
(a)



(b)

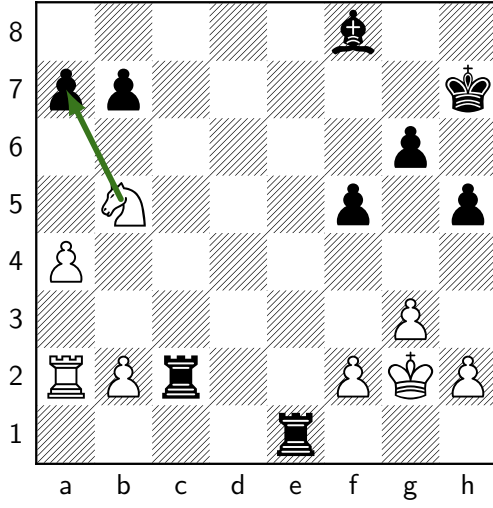


(c)

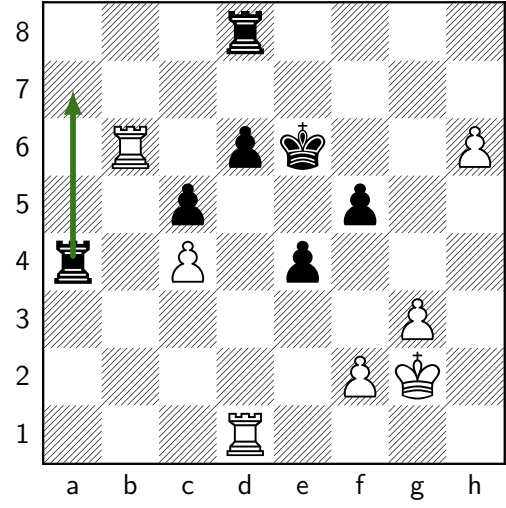


(d)

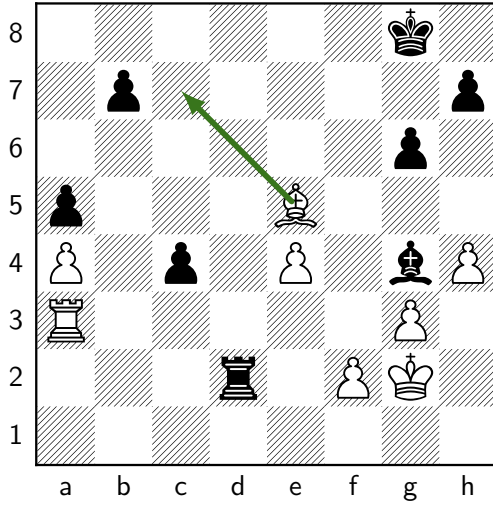
Figure 32: **Move 38, White: Nb5**. (a) Our position; Melanie plays Nb5 (c3b5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2096` (ply 83) [structural distance $d=43$, QAP obj=13]: the master played Rd7; under μ our Nc3 plays the role of that game’s Rf2 (a **cross-type** correspondence); cross-type roles: our Ra2 as their Nd6, our rc4 as their bg6; transferred move c3b5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1528` (ply 63) [structural distance $d=43$, QAP obj=12]: the master played Nf6; under μ our Nc3 plays the role of that game’s Nd4; cross-type roles: our pb7 as their Rd7, our Ra2 as their Pb6; transferred move c3b5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#2304` (ply 71) [structural distance $d=28$, QAP obj=27]: the master played Nd4; under μ our Nc3 plays the role of that game’s Nc3; cross-type roles: our re1 as their Pf6, our Pf2 as their bc4; transferred move c3d5. Synthesis: 2 retrieved analogs transferred to Nb5 (summed QAP objective 25, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



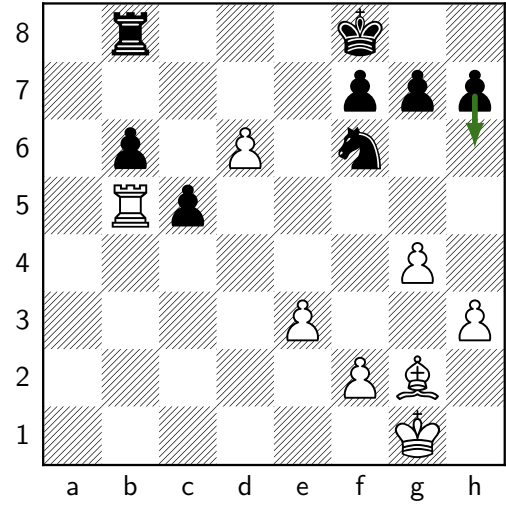
(a)



(b)



(c)



(d)

Figure 33: **Move 39, White: Nxa7.** (a) Our position; Melanie plays Nxa7 (b5a7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#848` (ply 93) [structural distance $d=38$, QAP obj=39]: the master played Ra7; under μ our Nb5 plays the role of that game’s Ra5 (a **cross-type** correspondence); cross-type roles: our re1 as their ph6, our bf8 as their rd1; transferred move b5a7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1948` (ply 80) [structural distance $d=38$, QAP obj=24]: the master played Bc7; under μ our Nb5 plays the role of that game’s Be5 (a **cross-type** correspondence); cross-type roles: our Pa4 as their Ra3, our re1 as their Pa4; transferred move b5a7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#2541` (ply 69) [structural distance $d=31$, QAP obj=64]: the master played h6; under μ our Pg3 plays the role of that game’s Ph2; cross-type roles: our re1 as their pe3, our bf8 as their ph3; transferred move g3g4. Synthesis: 2 retrieved analogs transferred to Nxa7 (summed QAP objective 63, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

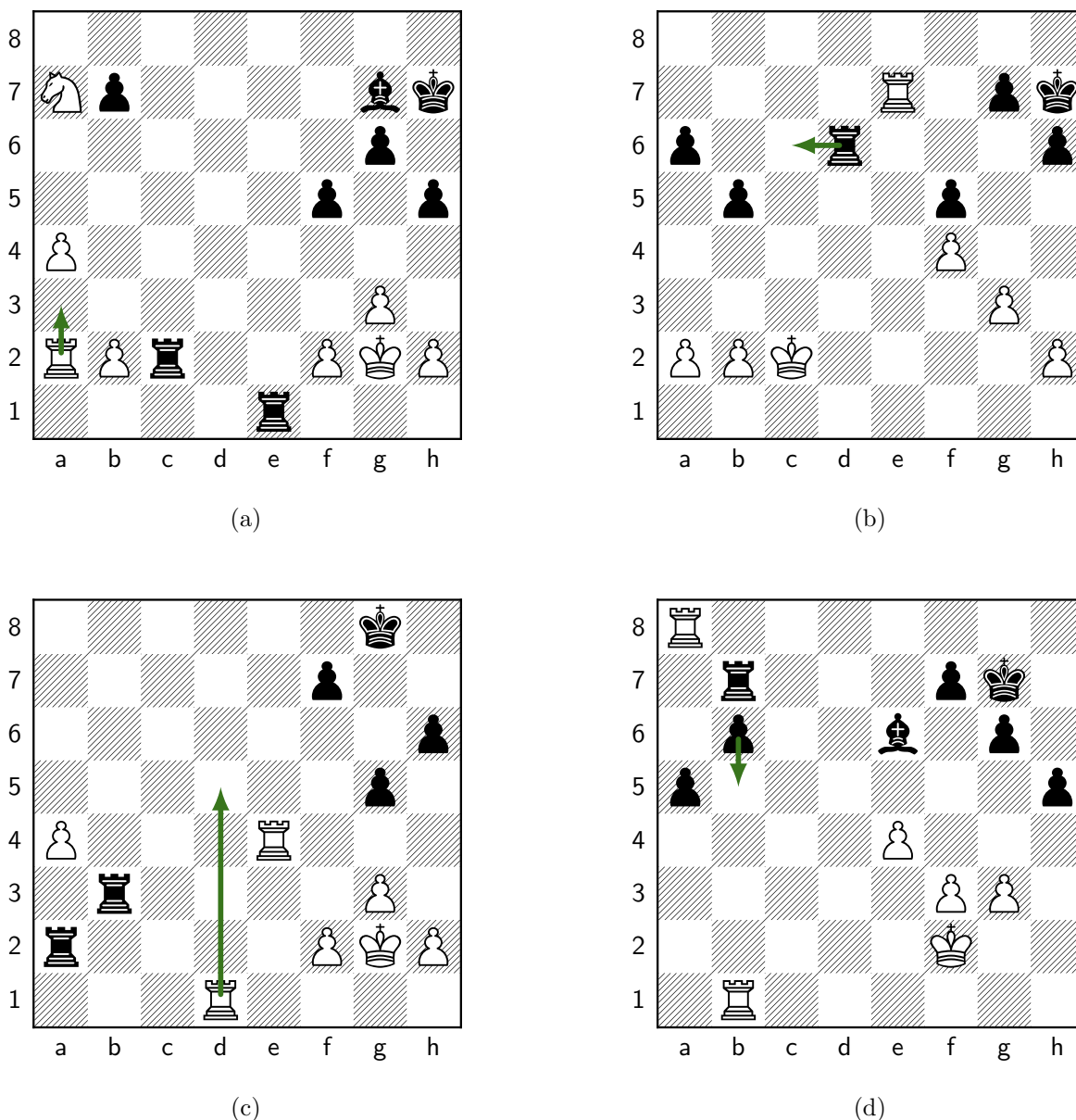
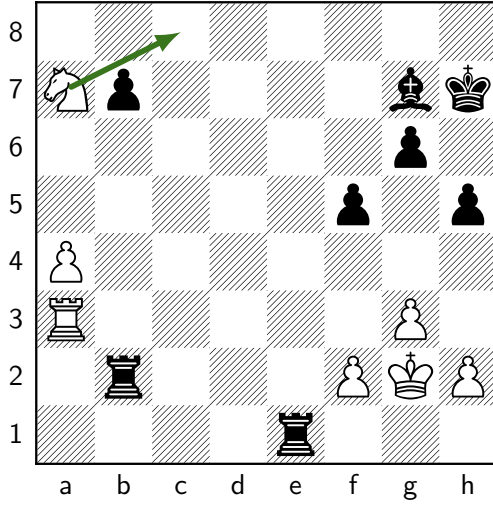
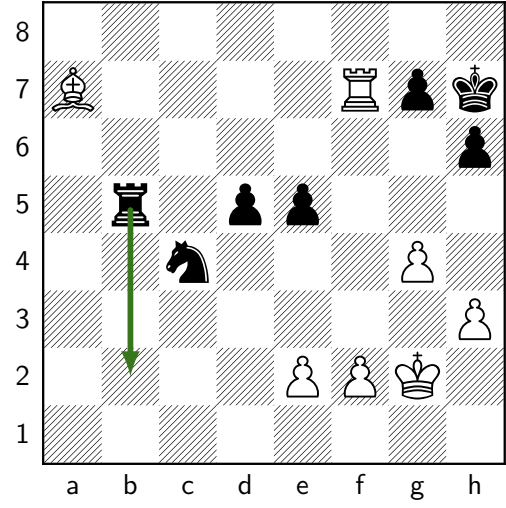


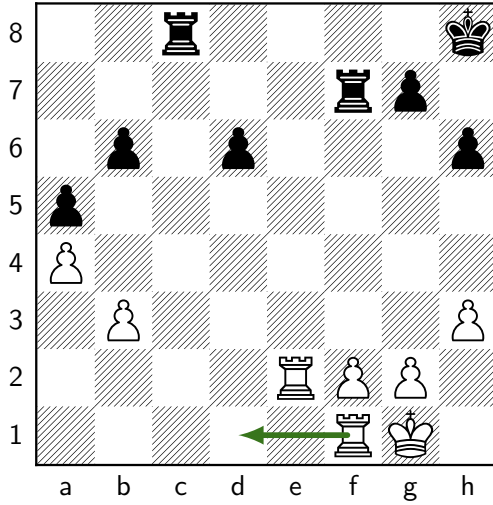
Figure 34: **Move 40, White: Ra3.** (a) Our position; Melanie plays Ra3 (a2a3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3279` (ply 83) [structural distance $d=34$, QAP obj=25]: the master played Rc6+; under μ our Ra2 plays the role of that game’s Rd3; cross-type roles: our Na7 as their Pf5, our re1 as their pa2; transferred move a2a3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3322` (ply 76) [structural distance $d=38$, QAP obj=17]: the master played Rd5; under μ our Ra2 plays the role of that game’s Rd1; cross-type roles: our Pa4 as their Re4, our bg7 as their pf7; transferred move a2a3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#2622` (ply 91) [structural distance $d=36$, QAP obj=13]: the master played b5; under μ our Ra2 plays the role of that game’s Pb3 (a **cross-type** correspondence); cross-type roles: our Na7 as their Rb7, our Pf2 as their Be6; transferred move a2a3 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Ra3 (summed QAP objective 79, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



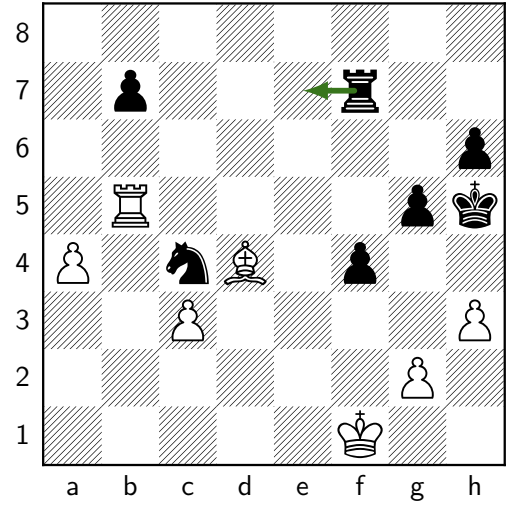
(a)



(b)

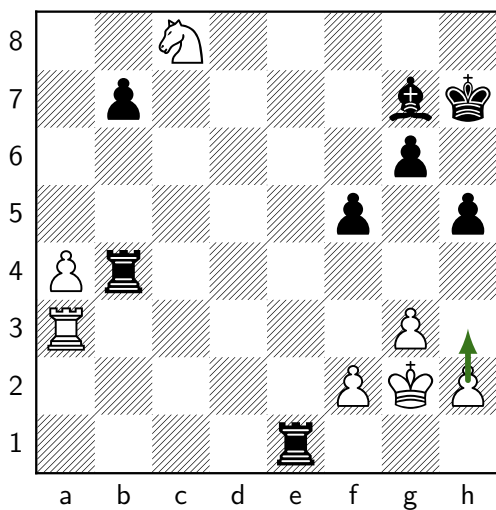


(c)

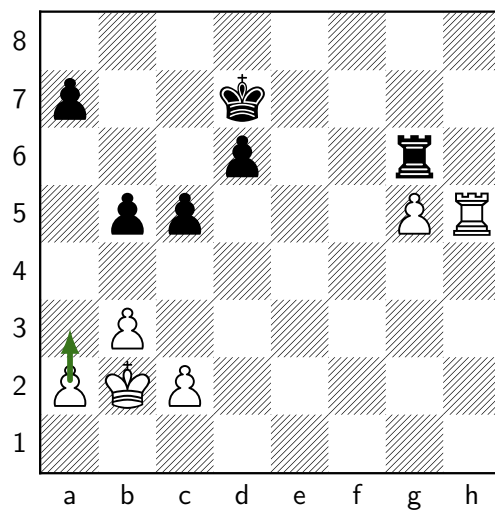


(d)

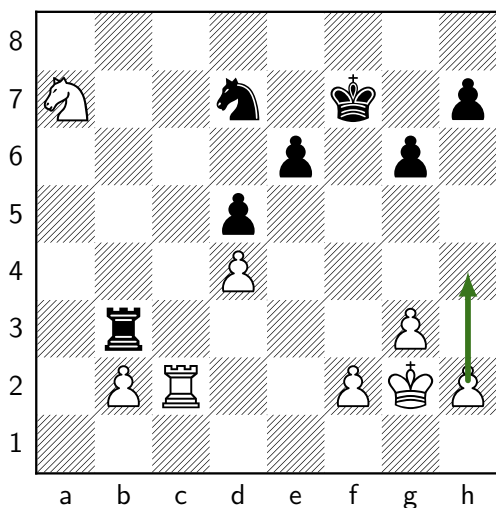
Figure 35: **Move 41, White: Nc8.** (a) Our position; Melanie plays Nc8 (a7c8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#398` (ply 67) [structural distance $d=28$, QAP obj=20]: the master played Rb2; under μ our Na7 plays the role of that game’s Rb4 (a **cross-type** correspondence); cross-type roles: our pb7 as their ba7, our re1 as their Pd5; transferred move a7c8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#1721` (ply 70) [structural distance $d=29$, QAP obj=39]: the master played Rd1; under μ our Pa4 plays the role of that game’s Rf1 (a **cross-type** correspondence); cross-type roles: our bg7 as their Re2, our re1 as their Ph3; transferred move a4a5. (d) Retrieved from `ccr1_top_p0000#3882` (ply 77) [structural distance $d=29$, QAP obj=34]: the master played Re7; under μ our Ra3 plays the role of that game’s Rf2; cross-type roles: our bg7 as their pa4, our pb7 as their bd4; transferred move a3d3. Synthesis: 1 retrieved analog transferred to Nc8 (summed QAP objective 20, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



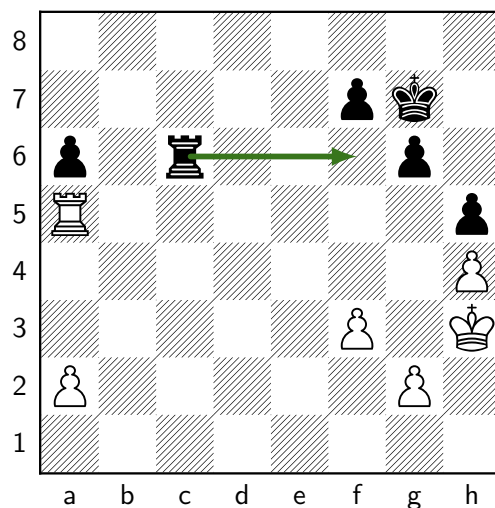
(a)



(b)

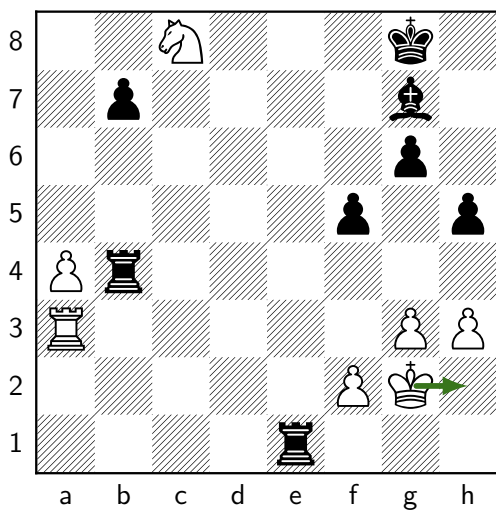


(c)

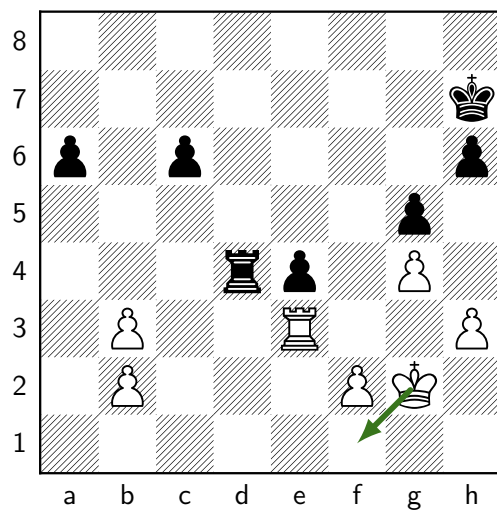


(d)

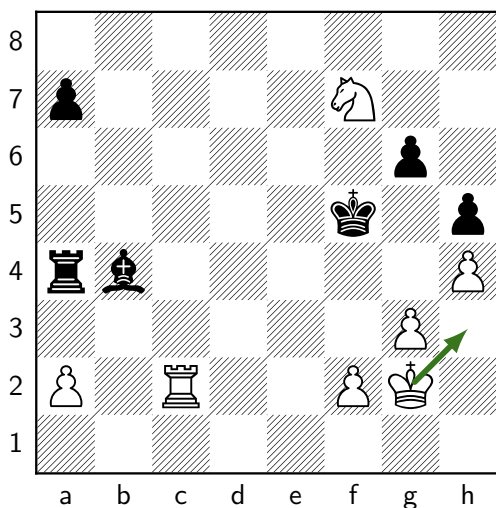
Figure 36: **Move 42, White: h3.** (a) Our position; Melanie plays **h3** (**h2h3**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#433` (ply 76) [structural distance $d=27$, QAP obj=26]: the master played **a3**; under μ our *Ph2* plays the role of that game’s *Pa2*; cross-type roles: our *re1* as their *pa7*; transferred move **h2h3** (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3862` (ply 70) [structural distance $d=16$, QAP obj=24]: the master played **h4**; under μ our *Ph2* plays the role of that game’s *Ph2*; cross-type roles: our *re1* as their *Pd4*, our *Nc8* as their *pd5*; transferred move **h2h4**. (d) Retrieved from `ccr1_top_p0001#3587` (ply 69) [structural distance $d=30$, QAP obj=23]: the master played **Rf6**; under μ our *Ra3* plays the role of that game’s *Rc3*; cross-type roles: our *bg7* as their *pg2*; transferred move **a3f3**. Synthesis: 1 retrieved analog transferred to **h3** (summed QAP objective 26, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



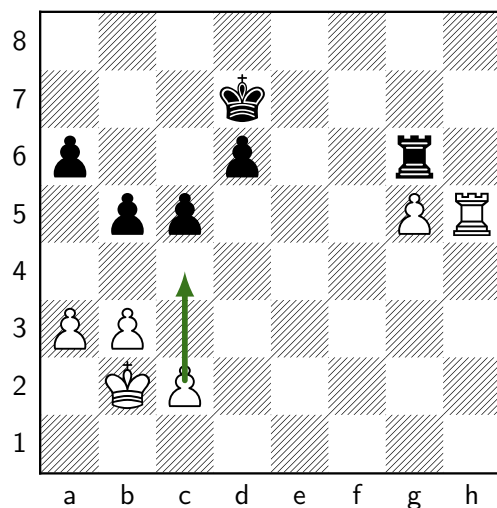
(a)



(b)

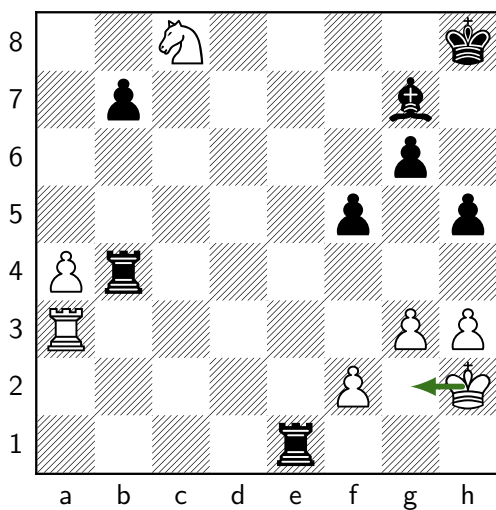


(c)

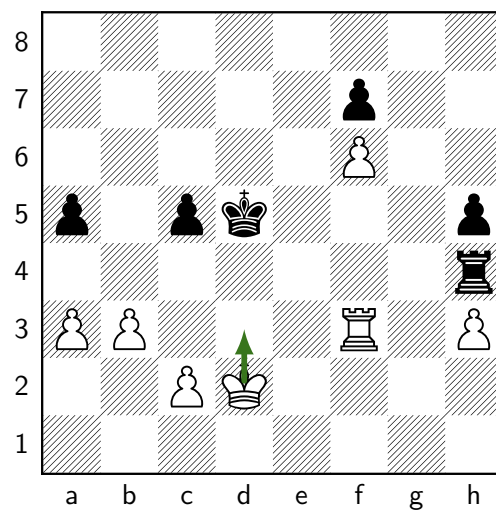


(d)

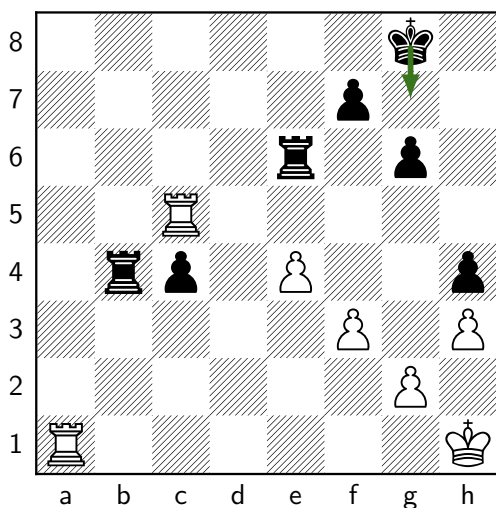
Figure 37: **Move 43, White: Kh2.** (a) Our position; Melanie plays Kh2 (g2h2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1641` (ply 86) [structural distance $d=32$, QAP obj=24]: the master played Kf1; under μ our Kg2 plays the role of that game’s Kg2; cross-type roles: our re1 as their pa6, our bg7 as their ph6; transferred move g2h2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3435` (ply 84) [structural distance $d=31$, QAP obj=13]: the master played Kh3; under μ our Kg2 plays the role of that game’s Kg2; cross-type roles: our pb7 as their bb4, our re1 as their pa7; transferred move g2h2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#433` (ply 78) [structural distance $d=33$, QAP obj=27]: the master played c4; under μ our Pf2 plays the role of that game’s Pc2; transferred move f2f4. Synthesis: 2 retrieved analogs transferred to Kh2 (summed QAP objective 37, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



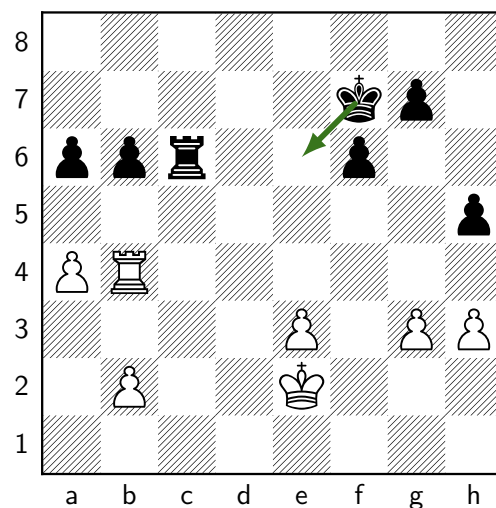
(a)



(b)

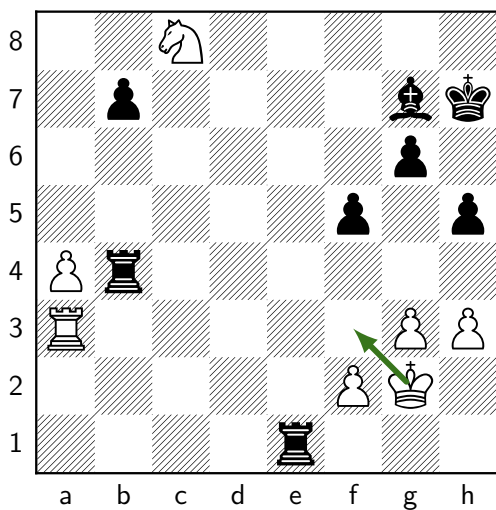


(c)

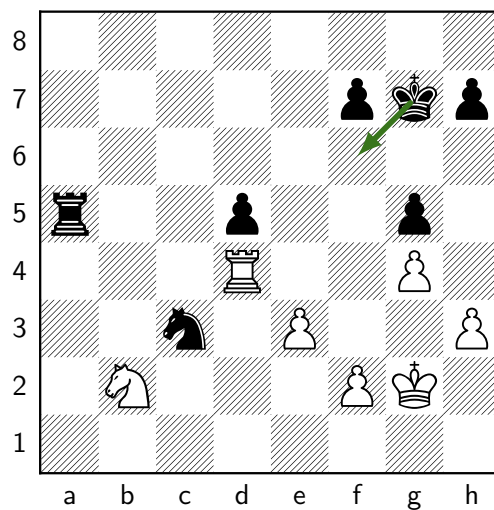


(d)

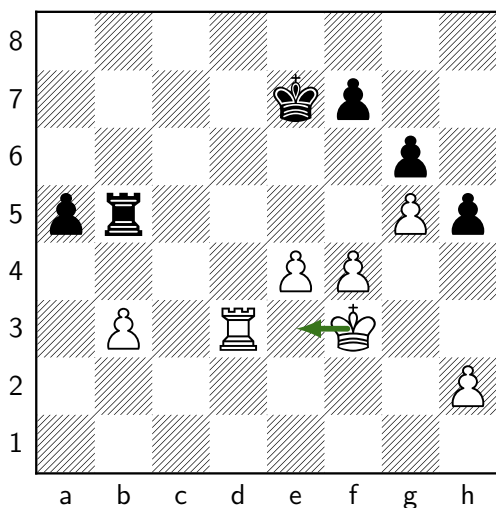
Figure 38: **Move 44, White: Kg2.** (a) Our position; Melanie plays Kg2 (**h2g2**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#1441` (ply 82) [structural distance $d=31$, QAP obj=23]: the master played Kd3; under μ our *Kh2* plays the role of that game’s *Kd2*; cross-type roles: our *rel* as their *Pf6*, our *Nc8* as their *pf7*; transferred move **h2g2 (endorses this move)**. (c) Retrieved from `ccrl_top_p0002#1157` (ply 77) [structural distance $d=32$, QAP obj=23]: the master played Kg7; under μ our *Kh2* plays the role of that game’s *Kg1*; cross-type roles: our *Nc8* as their *Pf7*; transferred move **h2g2 (endorses this move)**. (d) Retrieved from `ccrl_top_p0002#1145` (ply 77) [structural distance $d=31$, QAP obj=14]: the master played Ke6; under μ our *Kh2* plays the role of that game’s *Kf2*; cross-type roles: our *Nc8* as their *Pg7*, our *rel* as their *Pf6*; transferred move **h2g2 (endorses this move)**. Synthesis: 4 retrieved analogs transferred to Kg2 (summed QAP objective 73, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



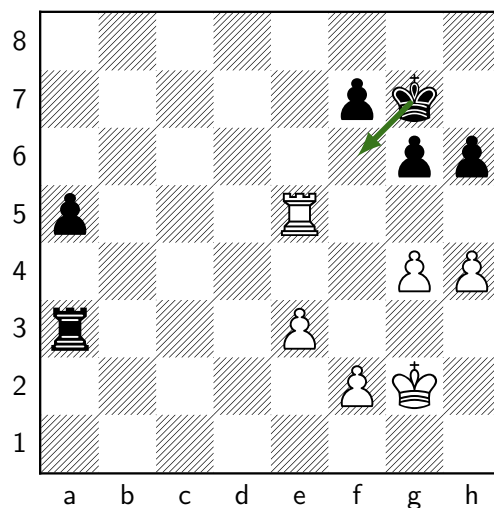
(a)



(b)

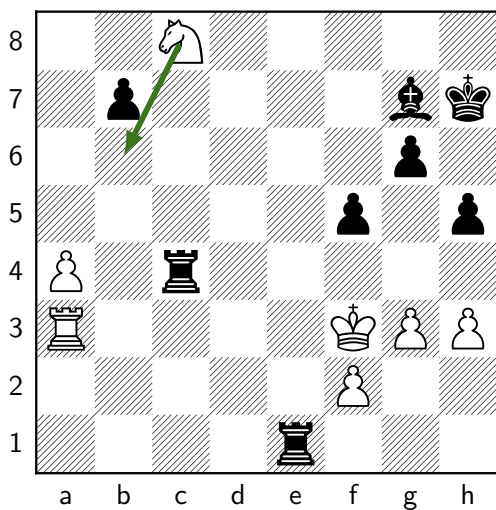


(c)

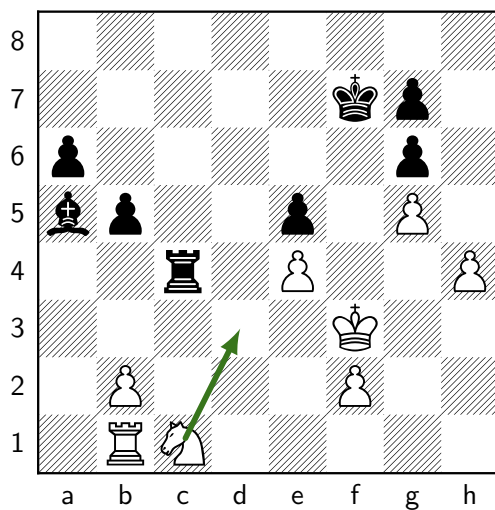


(d)

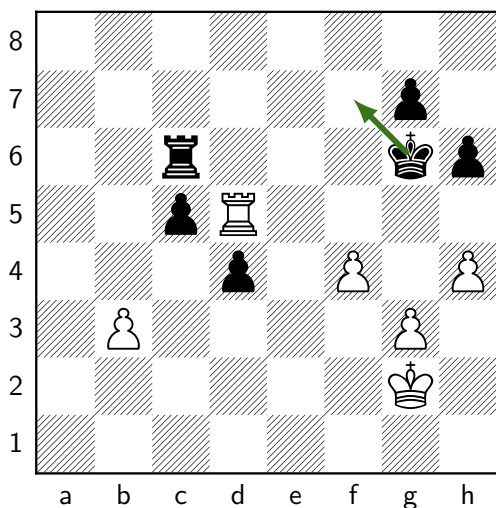
Figure 39: **Move 45, White: Kf3.** (a) Our position; Melanie plays Kf3 (g2f3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#475` (ply 75) [structural distance $d=29$, QAP obj=24]: the master played Kf6; under μ our Kg2 plays the role of that game’s Kg2; cross-type roles: our Ra3 as their Nc3; transferred move g2f3 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#3065` (ply 78) [structural distance $d=31$, QAP obj=13]: the master played Ke3; under μ our Kg2 plays the role of that game’s Kf3; cross-type roles: our re1 as their ph5; transferred move g2f3 (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#1146` (ply 87) [structural distance $d=29$, QAP obj=12]: the master played Kf6; under μ our Kg2 plays the role of that game’s Kg2; cross-type roles: our bg7 as their pf2; transferred move g2f3 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Kf3 (summed QAP objective 50, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



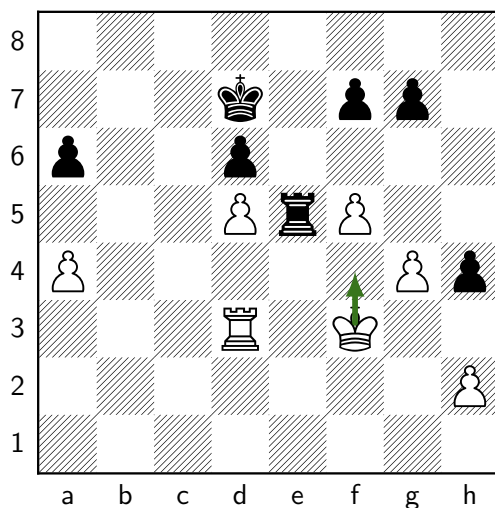
(a)



(b)

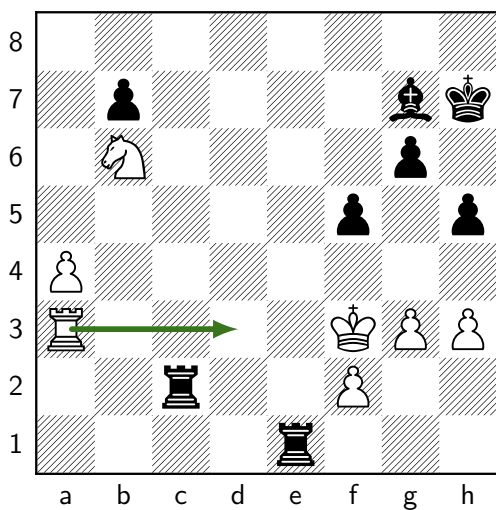


(c)

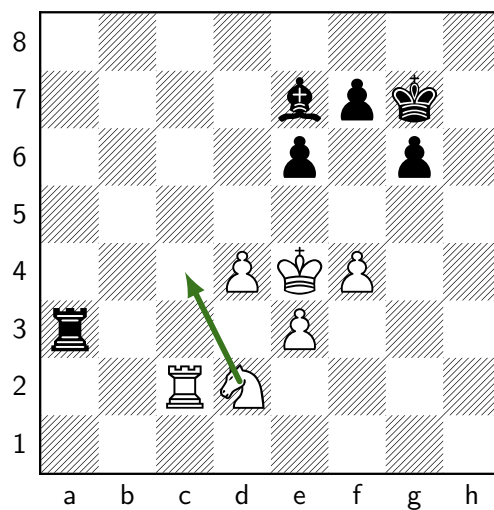


(d)

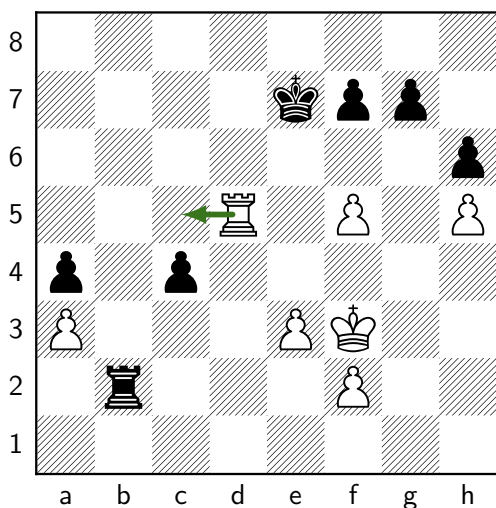
Figure 40: **Move 46, White: Nb6.** (a) Our position; Melanie plays Nb6 (c8b6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccl1.top_p0002#830` (ply 90) [structural distance $d=32$, QAP obj=28]: the master played Nd3; under μ our Nc8 plays the role of that game’s Nc1; cross-type roles: our pf5 as their ba5, our re1 as their pe5; transferred move c8b6 (**endorses this move**). (c) Retrieved from `ccl1.top_p0000#1306` (ply 91) [structural distance $d=26$, QAP obj=31]: the master played Kf7; under μ our Kf3 plays the role of that game’s Kg3; cross-type roles: our Nc8 as their Pd4; transferred move f3g2. (d) Retrieved from `ccl1.top_p0002#2316` (ply 86) [structural distance $d=35$, QAP obj=27]: the master played Kf4; under μ our Kf3 plays the role of that game’s Kf3; cross-type roles: our Nc8 as their Pd5, our Nc8 as their Pf5; transferred move f3g2. Synthesis: 1 retrieved analog transferred to Nb6 (summed QAP objective 28, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



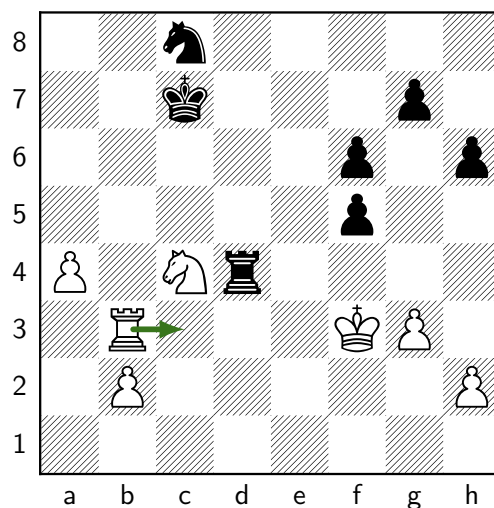
(a)



(b)



(c)



(d)

Figure 41: **Move 47, White: Rd3.** (a) Our position; Melanie plays Rd3 (a3d3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#853` (ply 100) [structural distance $d=32$, QAP obj=21]: the master played Nc4; under μ our *Ra3* plays the role of that game's *Nd2* (a **cross-type** correspondence); cross-type roles: our *Nb6* as their *Rc2*, our *re1* as their *be7*; transferred move a3d3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3188` (ply 82) [structural distance $d=33$, QAP obj=20]: the master played Rc5; under μ our *Ra3* plays the role of that game's *Rd5*; cross-type roles: our *Nb6* as their *Pa3*, our *bg7* as their *pc4*; transferred move a3d3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#1696` (ply 96) [structural distance $d=28$, QAP obj=3]: the master played Rc3; under μ our *Ra3* plays the role of that game's *Rb3*; cross-type roles: our *re1* as their *Pa4*, our *Pf2* as their *Nc4*; transferred move a3d3 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Rd3 (summed QAP objective 44, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

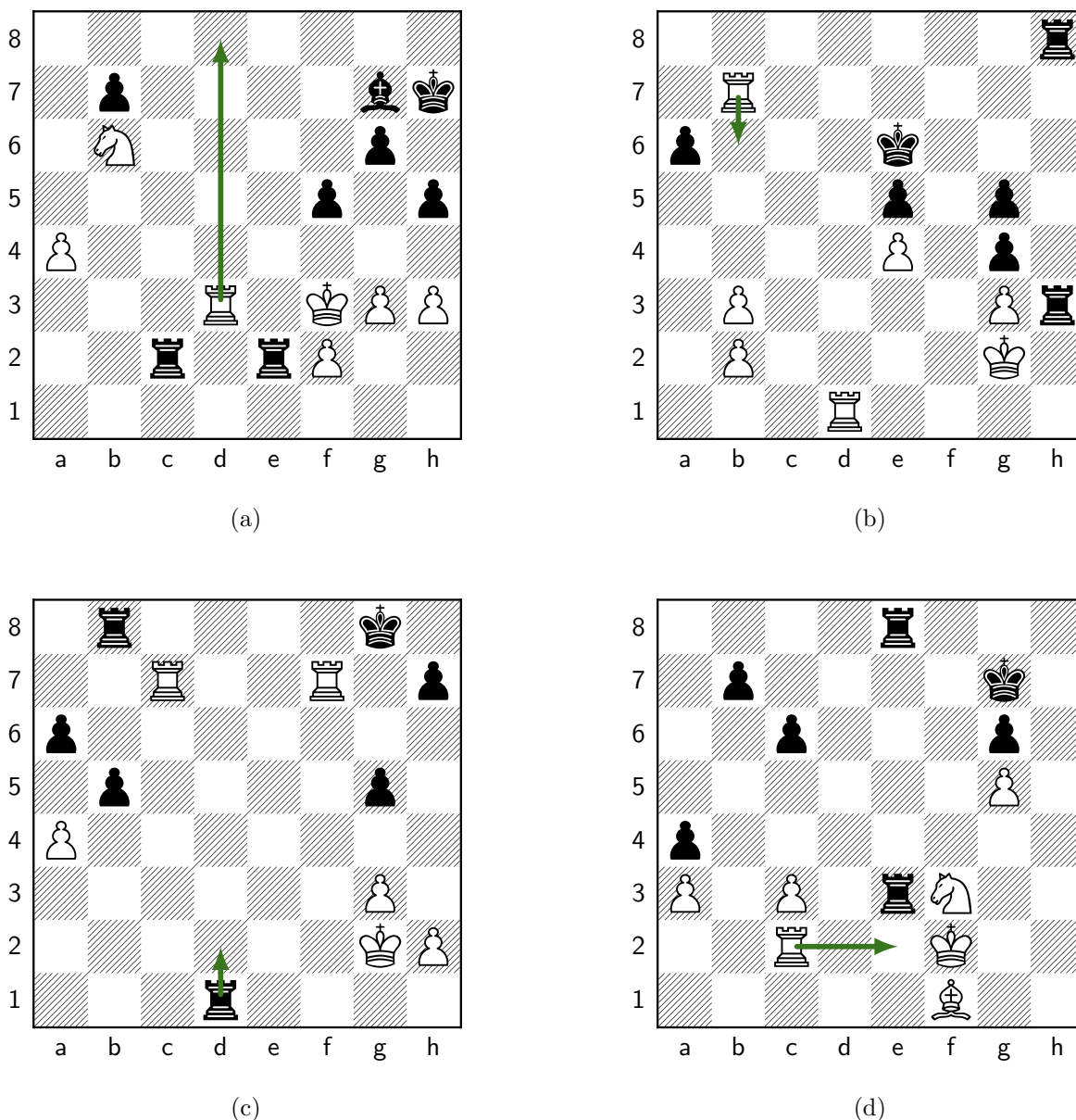
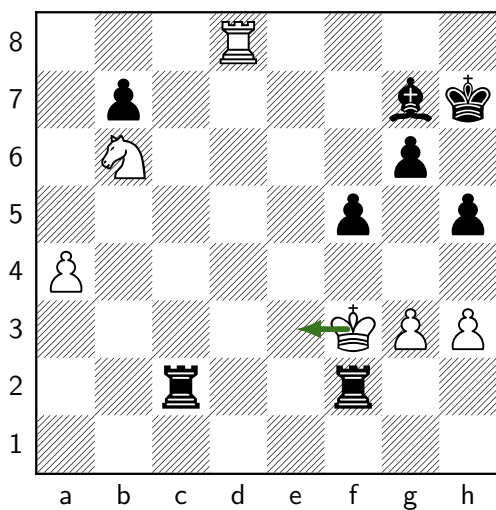
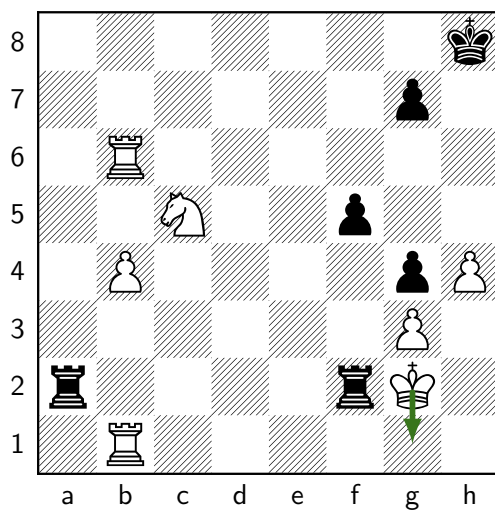


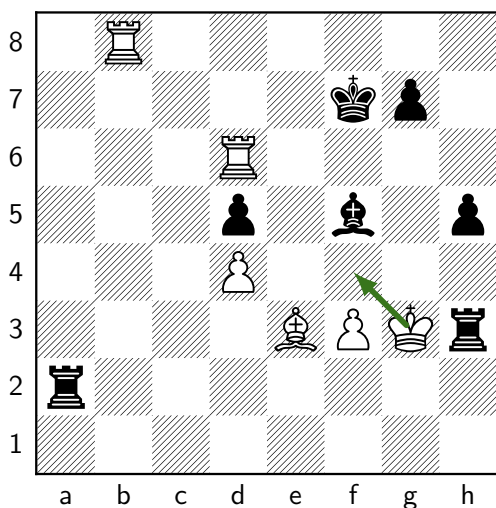
Figure 42: **Move 48, White: Rd8.** (a) Our position; Melanie plays Rd8 (d3d8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#372` (ply 92) [structural distance $d=38$, QAP obj=39]: the master played Rb6+; under μ our Rd3 plays the role of that game’s Rb7; cross-type roles: our Ph3 as their Rd1, our Nb6 as their Pb3; transferred move d3d8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2734` (ply 83) [structural distance $d=38$, QAP obj=40]: the master played Rd2+; under μ our Pa4 plays the role of that game’s Rd8 (a **cross-type** correspondence); cross-type roles: our Nb6 as their Pb5; transferred move a4a5. (d) Retrieved from `ccr1_top_p0001#3292` (ply 104) [structural distance $d=39$, QAP obj=30]: the master played Re2; under μ our Rd3 plays the role of that game’s Rc2; cross-type roles: our Pg3 as their Bf1, our Pf2 as their Nf3; transferred move d3e3. Synthesis: 2 retrieved analogs transferred to Rd8 (summed QAP objective 78, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



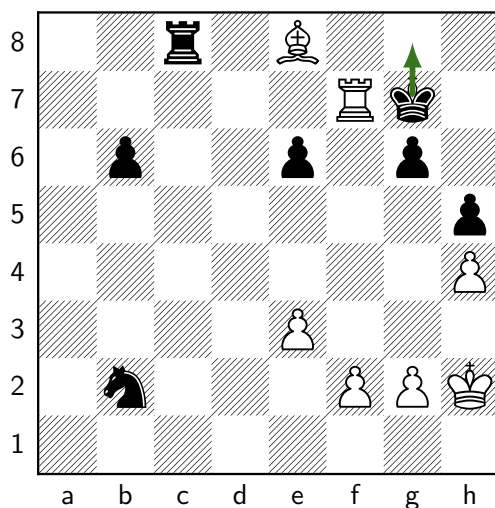
(a)



(b)

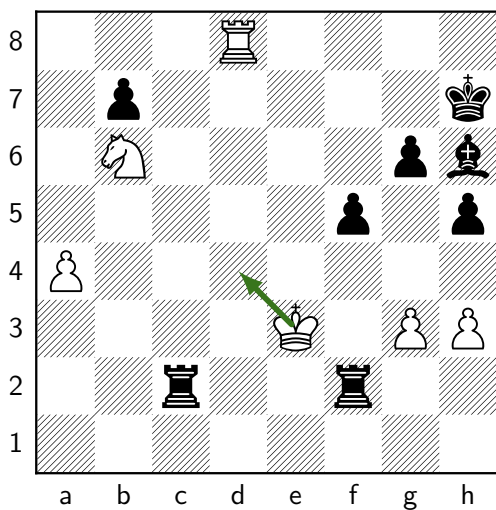


(c)

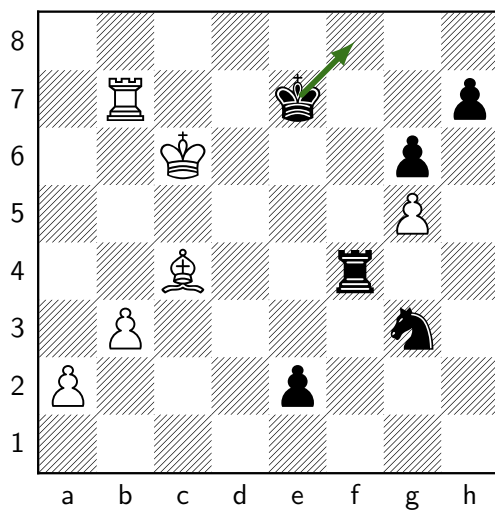


(d)

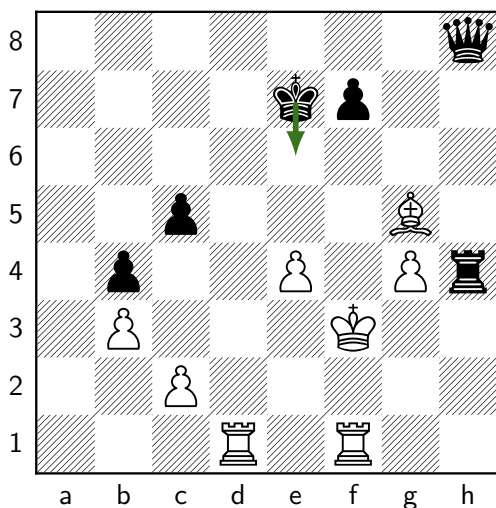
Figure 43: **Move 49, White: Ke3.** (a) Our position; Melanie plays Ke3 (f3e3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#340` (ply 88) [structural distance $d=36$, QAP obj=74]: the master played Kg1; under μ our Kf3 plays the role of that game's Kg2; cross-type roles: our Ph3 as their Rb1, our bg7 as their pg7; transferred move f3e3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2165` (ply 98) [structural distance $d=37$, QAP obj=74]: the master played Kf4; under μ our Kf3 plays the role of that game's Kg3; cross-type roles: our ph5 as their ra2, our pg6 as their Be3; transferred move f3e3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#2771` (ply 81) [structural distance $d=36$, QAP obj=70]: the master played Kg8; under μ our Kf3 plays the role of that game's Kg2; cross-type roles: our ph5 as their be8, our bg7 as their pe3; transferred move f3e3 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Ke3 (summed QAP objective 243, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



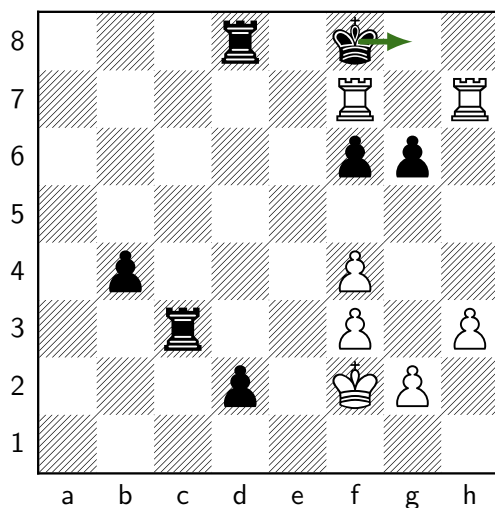
(a)



(b)



(c)



(d)

Figure 44: **Move 50, White: Kd4.** (a) Our position; Melanie plays Kd4 (e3d4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2772` (ply 93) [structural distance $d=38$, QAP obj=72]: the master played Kf8; under μ our Ke3 plays the role of that game's Ke2; cross-type roles: our bh6 as their rb7, our rc2 as their bc4; transferred move e3d3. (c) Retrieved from `ccr1_top_p0002#1721` (ply 91) [structural distance $d=39$, QAP obj=61]: the master played Ke6; under μ our Ke3 plays the role of that game's Ke2; cross-type roles: our Nb6 as their Qh8, our rf2 as their pg4; transferred move e3d3. (d) Retrieved from `ccr1_top_p0000#3362` (ply 113) [structural distance $d=41$, QAP obj=27]: the master played Kg8; under μ our Ke3 plays the role of that game's Kf1; cross-type roles: our pb7 as their Rd8, our bh6 as their rf7; transferred move e3d3. Synthesis: no retrieved analog transferred to Kd4 – the search selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.

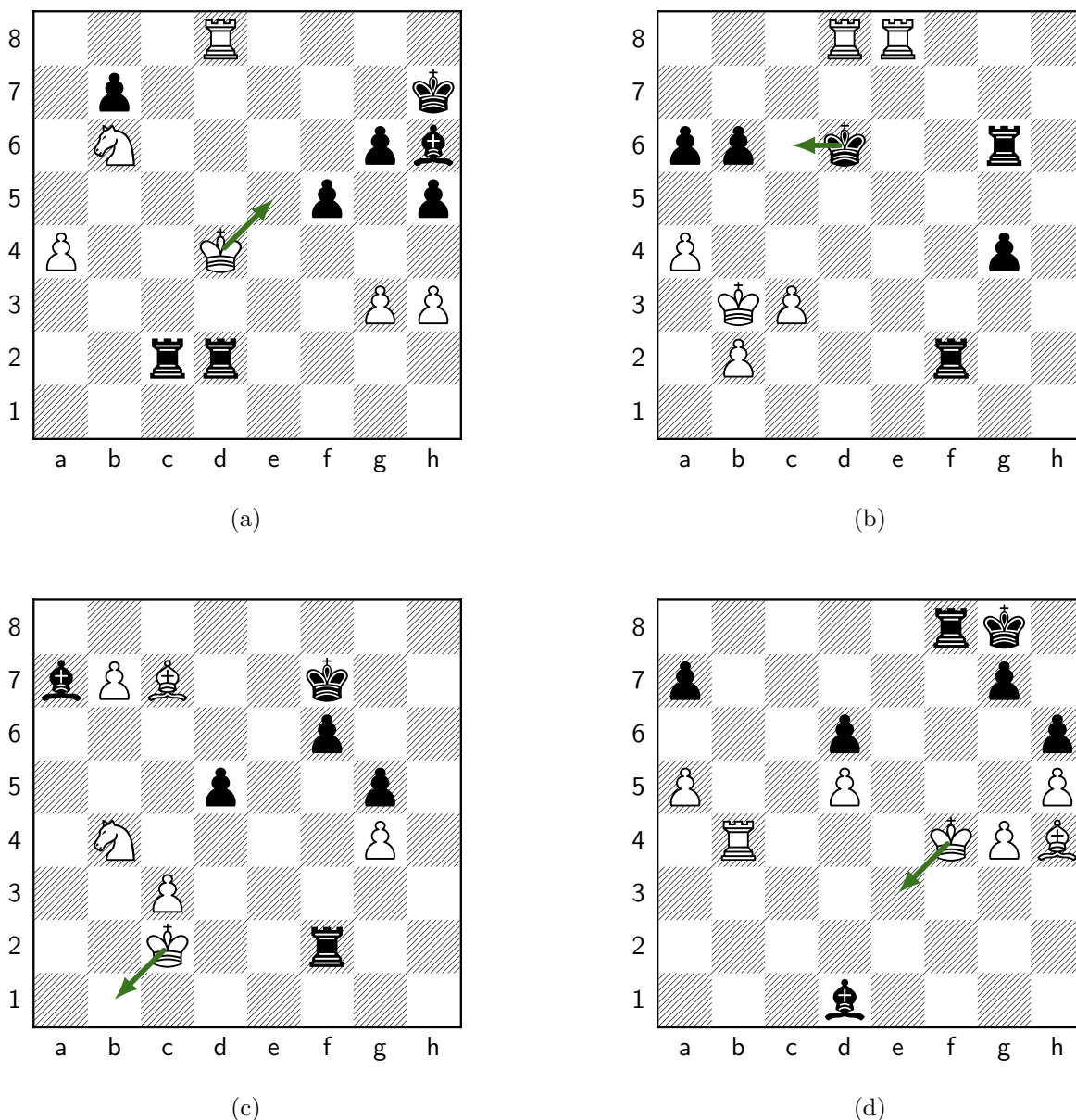
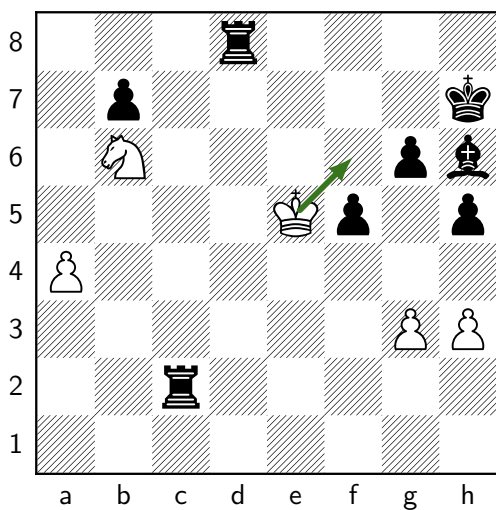
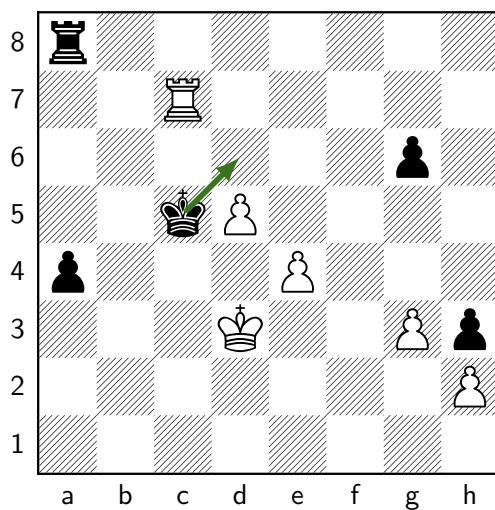


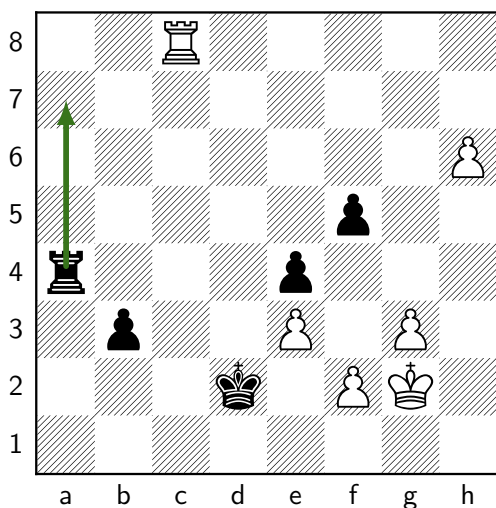
Figure 45: **Move 51, White: Ke5.** (a) Our position; Melanie plays Ke5 (d4e5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#405` (ply 117) [structural distance $d=34$, QAP obj=74]: the master played Kc6; under μ our Kd4 plays the role of that game’s Kd3; cross-type roles: our bh6 as their pc3, our Nb6 as their Rf2; transferred move d4e5 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#2709` (ply 114) [structural distance $d=40$, QAP obj=51]: the master played Kb1; under μ our Kd4 plays the role of that game’s Kc2; cross-type roles: our rc2 as their Pc3, our Rd8 as their Nb4; transferred move d4e5 (**endorses this move**). (d) Retrieved from `ccrl_top_p0003#991` (ply 84) [structural distance $d=40$, QAP obj=42]: the master played Ke3; under μ our Kd4 plays the role of that game’s Kf4; cross-type roles: our rc2 as their bd1, our bh6 as their Pg4; transferred move d4e5 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Ke5 (summed QAP objective 209, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



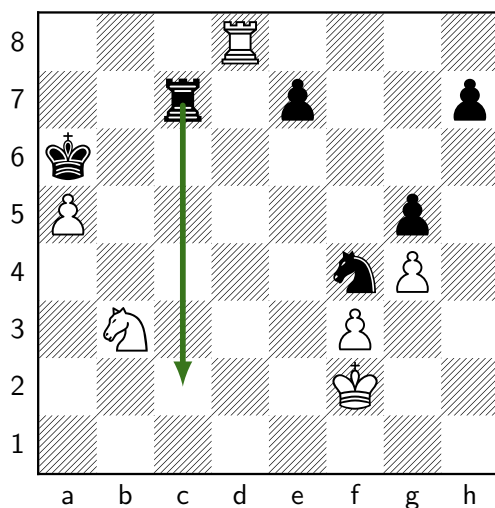
(a)



(b)

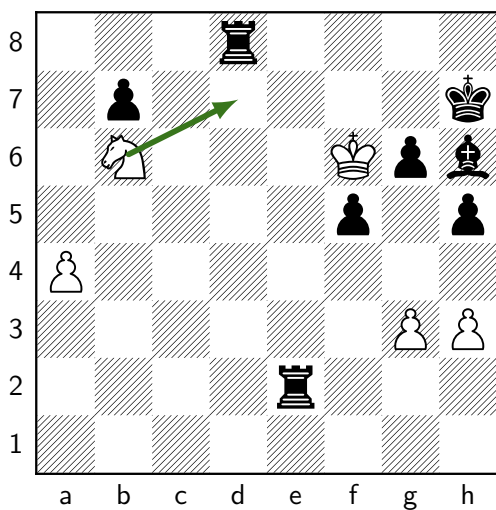


(c)

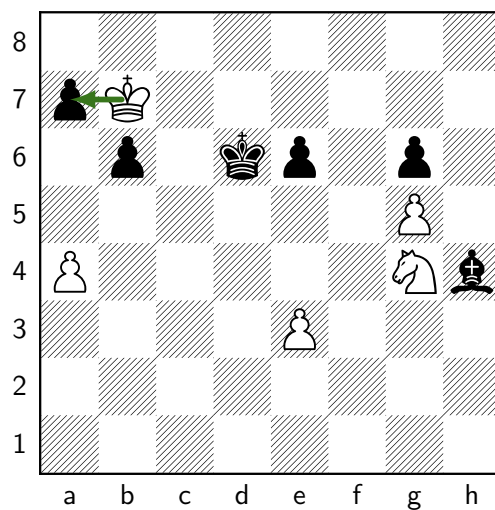


(d)

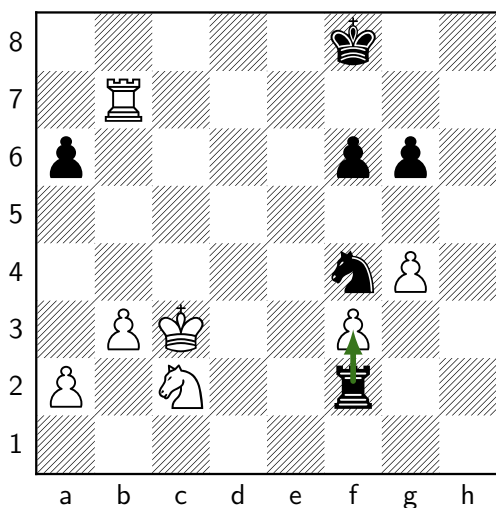
Figure 46: **Move 52, White: Kf6.** (a) Our position; Melanie plays Kf6 (e5f6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#3720` (ply 95) [structural distance $d=27$, QAP obj=4]: the master played Kd6; under μ our *Ke5* plays the role of that game’s *Kc4*; cross-type roles: our *Nb6* as their *Ra8*, our *rc2* as their *pg3*; transferred move e5f6 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#3674` (ply 109) [structural distance $d=23$, QAP obj=24]: the master played Ra7; under μ our *Ph3* plays the role of that game’s *Ra5* (a **cross-type** correspondence); cross-type roles: our *rc2* as their *ph6*, our *Nb6* as their *Pb3*; transferred move h3h4. (d) Retrieved from `ccrl_top_p0002#3865` (ply 113) [structural distance $d=27$, QAP obj=20]: the master played Rc2+; under μ our *Nb6* plays the role of that game’s *Rc2* (a **cross-type** correspondence); cross-type roles: our *pb7* as their *rd8*, our *rc2* as their *Ph7*; transferred move b6a8. Synthesis: 1 retrieved analog transferred to Kf6 (summed QAP objective 4, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



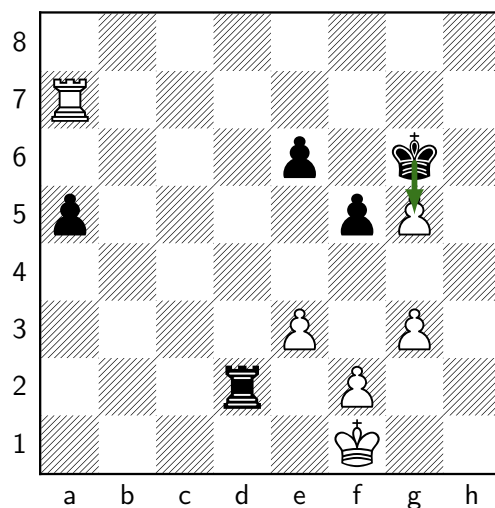
(a)



(b)

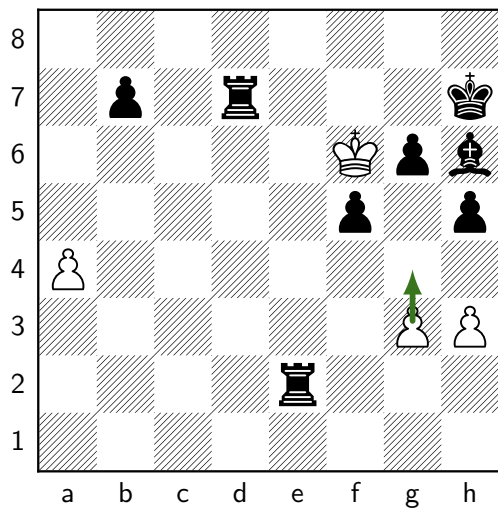


(c)

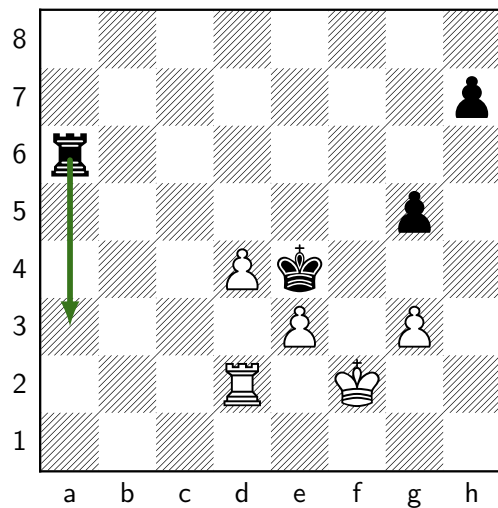


(d)

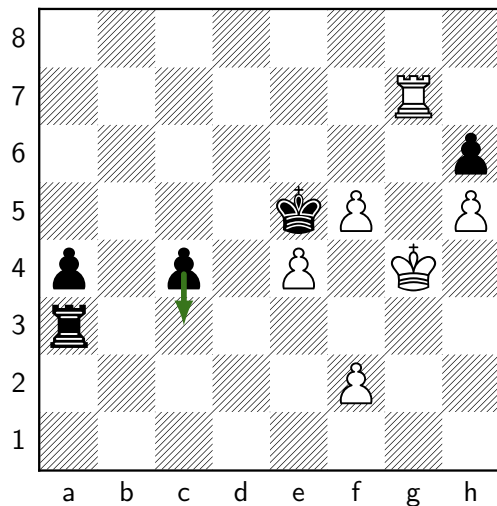
Figure 47: **Move 53, White: Nd7.** (a) Our position; Melanie plays Nd7 (b6d7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3976` (ply 114) [structural distance $d=23$, QAP obj=65]: the master played Kxa7; under μ our Kf6 plays the role of that game’s Kb7; cross-type roles: our re2 as their Ng4, our Kf6 as their bh4; transferred move f6f7. (c) Retrieved from `ccr1_top_p0003#2246` (ply 111) [structural distance $d=30$, QAP obj=38]: the master played Rxf3+; under μ our Kf6 plays the role of that game’s Rf7 (a **cross-type** correspondence); cross-type roles: our Nb6 as their Pa6, our Pa4 as their Nf4; transferred move f6f7. (d) Retrieved from `ccr1_top_p0003#822` (ply 89) [structural distance $d=29$, QAP obj=25]: the master played Kxg5; under μ our Kf6 plays the role of that game’s Kg3; cross-type roles: our Nb6 as their Pf5, our rd8 as their pg3; transferred move f6f7. Synthesis: no retrieved analog transferred to Nd7 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



(a)

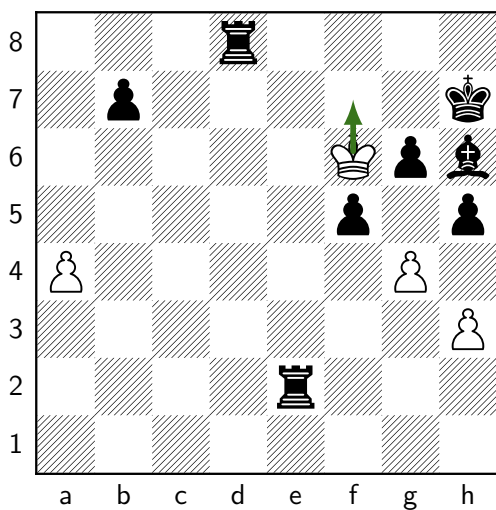


(b)

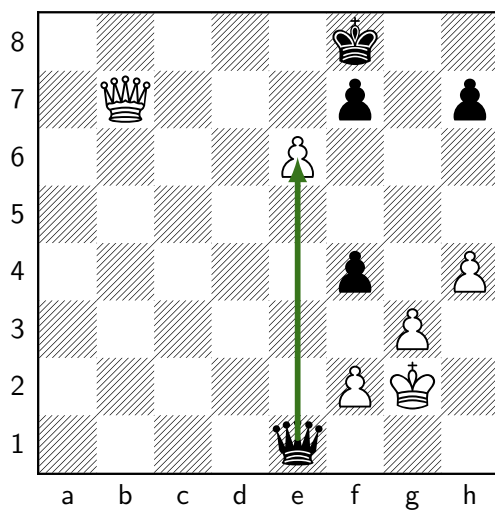


(c)

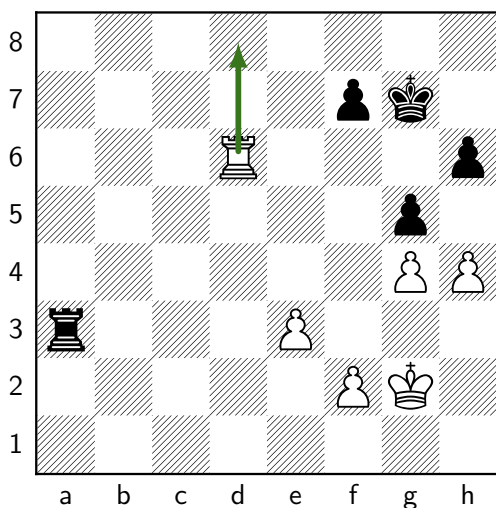
Figure 48: **Move 54, White: g4.** (a) Our position; Melanie plays g4 (g3g4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2073` (ply 93) [structural distance $d=22$, QAP obj=58]: the master played Ra3; under μ our Pg3 plays the role of that game’s Ra3 (a **cross-type** correspondence); cross-type roles: our re2 as their Ph7; transferred move g3g4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3188` (ply 93) [structural distance $d=26$, QAP obj=57]: the master played c3; under μ our Pa4 plays the role of that game’s Pc5; cross-type roles: our re2 as their Pa4, our bh6 as their Ra3; transferred move a4a5. Synthesis: 1 retrieved analog transferred to g4 (summed QAP objective 58, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



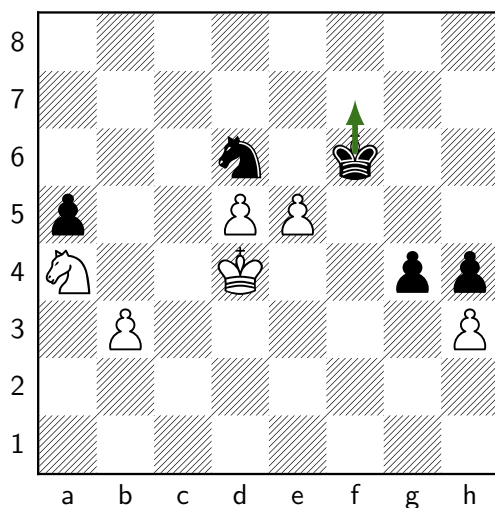
(a)



(b)



(c)



(d)

Figure 49: **Move 55, White: Kf7.** (a) Our position; Melanie plays Kf7 (f6f7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2087` (ply 97) [structural distance $d=37$, QAP obj=96]: the master played Qxe6; under μ our Kf6 plays the role of that game’s Qe8 (a **cross-type** correspondence); cross-type roles: our ph5 as their qb7, our re2 as their Ph7; transferred move f6f7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#1839` (ply 116) [structural distance $d=37$, QAP obj=42]: the master played Rd8; under μ our Kf6 plays the role of that game’s Rd6 (a **cross-type** correspondence); cross-type roles: our ph5 as their ra3, our re2 as their Pg4; transferred move f6f7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#483` (ply 119) [structural distance $d=38$, QAP obj=4]: the master played Kf7; under μ our Kf6 plays the role of that game’s Kf3; cross-type roles: our pf5 as their Nd6, our bh6 as their pe5; transferred move f6f7 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Kf7 (summed QAP objective 151, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.